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SADULLAEV LAZIZBEK OMONBOY UGLI

RESEARCH OF DYNAMIC SYSTEMS MODELLING CYCLIC PROCESSES

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Head of department «Mathematical engineering »:

_____ A. A. Reyimberganov

«____»_____2022 y.

Supervisor:

_____ PhD. Begimov M. O.

Reviewer:

_____ c.f.m.s. Reyimberganov A.A.

Urgench city – 2022

Annotatsiya

Ushbu dissertatsiyada siklik jarayonlarni modellashtiruvchi dinamik tizimlar tadbiqlari o‘rganilgan. Asosan, issiqlik almashinuvchilarida sodir bo‘ladigan siklik issiqlik almashinuv jarayonlarini modellashtirish ushbu issiqlik almashinuvi jarayonini boshqarish muammosini hal qilishga xizmat qiladi. Ishda aylanuvchi regenerativ havo isitgichi, noan’anaviy geometrik shaklga, ya’ni kesilgan konus ko‘rinishidagi holi o‘rganilgan. Jarayonni ifodalovchi modelning sonli usulda yechilgan. Ishlab chiqilgan algoritmlar asosida Python dasturlash tilida kompyuter dasturi ishlab chiqilgan.

Annotation

This dissertation investigates the application of dynamic systems that model cyclic processes. Basically, it serves to solve the problem of managing this heat exchange process by modeling cyclic heat exchange processes that take place in heat exchangers. We have studied the case of a rotating regenerative air heater with a different geometric shape, in the form of a truncated cone, where air flows in and out from 2 places on one side. A mathematical model and an algorithm corresponding to the model has been created. Following this, a numerical calculation of this mathematical model has been carried out. A computer program based on the algorithm has been developed in python programming language.

Аннотация

В данной диссертации исследуется применение динамических систем, моделирующих циклические процессы. В основном он служит для решения задачи управления этим теплообменным процессом путем моделирования циклических теплообменных процессов, протекающих в теплообменниках. Нами был изучен случай вращающегося регенеративного воздухонагревателя другой геометрической формы, в виде усеченного конуса, в который воздух входит и выходит из 2-х мест с

одной стороны. Создана математическая модель и алгоритм, соответствующий модели. После этого был проведен численный расчет данной математической модели. На основе алгоритма разработана компьютерная программа на языке программирования Python.

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INTRODUCTION

The actuality(relevance) and demand of the theme of the dissertation. These days a great deal of attention is being paid to the implementation of scientific research in our country. That is the development of practical work and the intensification of the practical application of scientific work is on demand. The requirements for the application of the results of scientific work in the real sector of the economy in scientific-fundamental, scientific-innovative projects, announced by the Ministry of Support of Innovative Ideas of the Republic of Uzbekistan, are very strong and such work is supported widely in the country.

This dissertation focuses on the application of dynamic systems that model cyclic processes. Basically, it serves to solve the problem of managing this heat exchange process by modeling cyclic heat exchange processes that take place in heat exchangers and by studying the research properties of these models. To be more precise, at present, the production of electricity in our country is mainly the responsibility of thermal power plants. Other types of power sources, nuclear power plants, solar power plants have not yet started yielding electricity. The main source of electricity is thermal power plants. One of the most important equipment of thermal power plants is a heat exchange apparatus called a rotating regenerative air heater. An example of a cyclic process is the heat exchange that takes place in this apparatus. Therefore, we took our dissertation as an example of the heat exchange process in a cyclic dynamic system as a cyclic dynamic regenerative air heater. The dissertation is devoted to the application of heat exchange in a heat regenerative circulating air heater.

The comments on the literatures of the relevant research topic. To date heat exchange processes in such heat exchangers have been studied by many researchers which is highlighted in the first chapter of the dissertation. These researchers mostly used partial derivative equations to model the heat exchange process in air preheaters, but models represented by such equations

can face several problems when applied to a real process. The latest and most modern literature on modeling rotational cyclic heat exchange processes of such mathematical models is the book of Yu. A. Kirsanov. Basically, these models are expressed and applied using partial derivative differential equations. As a result of certain restrictions, an attempt was made to ensure the adequacy of the model.

Recently, in the laboratory of dynamic systems and their applications of the Institute of Mathematics, which deals with this process in Uzbekistan, under the leadership of Academician A. Azamov, the work on the implementation of this process has been carried out for several years. Academician A. Azamov proposed a new approach to the creation of a mathematical model of a rotating regenerative heater. This approach involves the development and application of a mathematical model of the process of sampling in the form of discrete dynamic systems of heat exchange in a cyclic regenerative air heater. The works of academicians A. Azamov and M. Begimov have been published in the Ural Mathematical Journal and a number of other journals. So, here is a mathematical model of the heat exchange process in a conventional rotary regenerative air heater. Discrete dynamic system of heat exchange in a traditional cylindrical heat exchanger, and a number of properties of this mathematical model are revealed and applied in practice.

Scientific novelty of research work. We have studied the case of a rotating regenerative air heater with a different geometric shape, in the form of a truncated cone, where air flows in and out from 2 places on one side. In this case, the purpose of the conical shape of the heating device is to increase its efficiency. In my master's dissertation, two types of heat exchangers are studied, the first is a mathematical model of the heat exchange process in a truncated cone-shaped heat exchanger. The second is the air comes in from two places on one side which is on the top, and the air comes out from the other side which is on the bottom, so there are 4 flows here. To facilitate the operation of this

regenerative air heater, it is possible to cite a number of works currently devoted to the modification of geometric structures to increase its efficiency. To change the geometric shape of traditional heat exchange processes today, that is, to create the current developed conical heat exchanger.

Scientific and practical significance of the research results. The reason why the geometric shape of the heat exchanger is in the shape of a truncated cone is that if the truncated cone is placed in the shape of a truncated cone, the base is upside down, the small base is at the bottom. Now, when the air coming in from the top under a certain pressure goes down, the base is small, so the pressure increases and leaves the heat exchanger faster, and the hot air in this part heats up less, the hot air loses its energy less. On the other hand, a pressure of the air decreases constantly. As a result, such a design was developed to prevent corrosion during heat exchange. Obtained result will increase the economic efficiency of thermal power plants

Research methods. What we will see in my dissertation is the history of other scientists who used PDEs and numerical methods for modelling heat exchange process in air preheaters, the construction of the mathematical model which was created by Abdulla Azamov using discrete dynamic systems, development of computer simulation of the constructed model using algorithms and programming languages.

Research problems and its assumptions. We need to create a mathematical model and create an algorithm based on discrete dynamic systems and develop a computer program based on this algorithm. In that visualization, this computer program must numerically show how much air enters and leaves at one temperature. The same thing applies to the process of passing a truncated cone and a four-way air flow. The urgency of the topic is that the requirements for the implementation of the results of scientific work in practice are very high. Here we take a process in the real sector of the economy of the Republic, create a mathematical model of this process and provide the necessary guidelines for

the implementation of the results of the study of the properties of this mathematical model. Our article is submitted for publication in the Ural Mathematical Journal.

The practical results of the study. A mathematical model was created and a numerical calculation of this mathematical model was carried out. Besides, an algorithm and corresponding computer program were created.

The purpose of the dissertation is to create a mathematical model of the process of heat exchange in circulating air heaters in thermal power plants and to create a computer program that performs numerical calculations.

The subject of the dissertation:

Mathematical model, application of methods of the theory of dynamic systems, application of computer software package and programming language, Python.

The object of the dissertation:

Rotating truncated cone and a four-stream heat exchanger, air preheater, rotating air preheater in traditional(cylindrical) shape.

The structure of the dissertation. The dissertation consists of an introduction, three chapters, a conclusion and a list of references. The first chapter is devoted to the analysis of existing mathematical models of this process, where the existing mathematical models of the heat exchange process in regenerative air heaters in the traditional form are studied and compared with each other, in short it is history part. The second chapter describes how the heat exchange process in this heat exchanger takes place. The second paragraph describes a mathematical model in the form of discrete dynamic systems. The third chapter is devoted to the creation of computer programs of mathematical models created in the second chapter. Here we have modeled our process in the COMSOL Multiphysics computer software package, which is now widely used in modern computer technology. Here we have used the features of the ready-

made software package. Finally, in the last paragraph of the third chapter, we wrote the algorithm and the program code corresponding to this algorithm. At the end of the dissertation there is a list of references, thank you for your attention.

CHAPTER I. CYCLIC HEAT PROCESSES IN HEAT EXCHANGERS

§1.1. Thermal processes in the regenerator

The heat transfer process in the regenerator is carried out cyclically. The cycle consists of two operating periods - heating and cooling - and may also include idle periods formed when switching from one operating period (heating or cooling nozzle) to another. During the heating period, the nozzle perceives excess internal energy (heat) from the hot coolant. During the cooling period, the energy accumulated by the nozzle is given to the cold coolant washing it, and so on. Thus, heat transfer is carried out with the participation of three substances: two heat carriers - hot and cold - and a heat-storing nozzle.

Due to the cyclic nature of heat transfer, the processes of heat exchange of the packing with heat carriers and the transfer of energy by thermal conduction of the packing are non-stationary. The consequence of the non-stationary nature of the processes is that the heat transfer capacity of the regenerator, in contrast to the heat exchanger, depends not only on the heat transfer coefficients of the heat carriers, the temperature difference between the heat carriers, the surface area of the heat exchanger, the thermal conductivity coefficient and the wall thickness, but also on mass, density and specific heat capacity of the packing, on the duration of individual periods of the cycle.

In the regenerator, the heat transfer process is accompanied by mass transfer processes, since the hot coolant always contains vapours of certain substances (water, sulfuric acid, etc.), the condensation temperature (dew point) of which can be quite high. When the dew point of a substance reaches the temperature of the nozzle, condensation and deposition of this substance begins on the surface of the latter. If the nozzle material is not resistant to this substance, then corrosion occurs, which reduces the service life and heating surfaces of the RAPH(Rotating-plate of regenerative Air Pre Heater), as well as the heat transfer capacity. To take measures to counteract corrosion (in the RAPH of power boilers, gas turbine plants) or clogging of the tract (corrosion products in the RAPH of boilers, frost in air-cooling machines), it is necessary

to have detailed and accurate information on the development of temperature fields in the nozzle in each from cycle periods. Information about temperature fields is also necessary in the case of high-temperature regenerators for metallurgical furnaces, gas turbine engines, etc., where there is a danger of exceeding the maximum allowable temperature for the packing material. Since direct measurements of the packing temperature, especially in rotating regenerators, are difficult, the only way to obtain information about temperature fields is thermal calculation using a mathematical model of the regenerator.

1.1.1. Estimation of the heat load of the regenerator

Known methods for thermal calculation of a regenerator [1, 2–12] formally differ little from methods for calculating a recuperator, since they are based on the same heat balance and heat transfer equations.

Assuming the temperatures of heat carriers and packing to be constant and equal to some values averaged over a period, the heat transfer equations for cold and hot heat carriers are presented as:

$$Q_0 = \alpha_0 F_{w,0}(\overline{t_0} - \overline{t_{w,0}}), \quad Q_1 = \alpha_1 F_{w,1}(\overline{t_{f,1}} - \overline{t_{w,1}}) \tag{1.1.1}$$

Where Q_0 - is the heat flux perceived by cold air, W; Q_1 - the same, given off by hot gas, W; α_j – coefficients of heat transfer of heat carriers with the packing surface in the j-th period, W/(m^2*K); j = 0 for the cold period and j = 1 for the hot one; F_j - is the surface area of the packing washed by the coolant in the j-th period, m^2; $\bar{t}_{w,j}$ - is the packing temperature averaged over the j-th period, K; $\bar{t}_{f,j}$ - is the coolant temperature averaged over the path and over the period in the j-th period, K.

In steady state $Q_0 = Q_1 = Q.$ The heat flux transferred from the hot coolant to the cold coolant is determined by the heat transfer equation

$$Q = kF_w \overline{\Delta t}_f \tag{1.1.2}$$

where $F_w = F_{w,0} = F_{w,1}$ is the area of the entire surface of the packing, m^2; $\overline{\Delta t}_f$ - average logarithmic temperature difference between heat carriers, K; k is the heat transfer coefficient, W/(m^2 *K).

Heat balance equations for heat carriers

$$Q = G_{f,0} c_{p,0} (t''_{f,0} - t'_{f,0}),\ Q = G_{f,1} c_{p,1} (t'_{f,1} - t''_{f,1}) \qquad (1.1.3)$$

make it possible to calculate the temperatures of heat carriers at the outlet of the regenerator $t''_{f,j}$, (at known inlet temperatures $t'_{f,j}$) and thermal.

Regenerator efficiency [7]

$$\eta_t = \frac{W_{f,0}(t''_{f,0} - t'_{f,0})}{W_{f,\min}(t'_{f,1} - t'_{f,0})} = \frac{W_{f,1}(t'_{f,1} - t''_{f,1})}{W_{f,\min}(t'_{f,1} - t'_{f,0})} \qquad (1.1.4)$$

In relations (2.1.3) and (2.1.4) $G_{f,j}$ is the mass flow rate of the coolant in the j-th period, kg/s; $c_{p,j}$ - specific isobaric heat capacity of the coolant, J / (kg*K); $W_{f,j} = G_{f,j} c_{p,j}$ is the water equivalent of the coolant in the j-th period, W/K; $W_{f,\min}$ - minimum value .

In those cases, when the water equivalents of heat carriers are the same, the thermal efficiency coincides with the coefficient (degree) of regeneration.

Equations (1.1.1) - (1.1.4) do not differ in form from similar equations for the heat exchanger. However, there are significant internal differences between them, namely:

1. In a recuperator, F_w is understood as the area of one of the sides of the wall, and not both, as in the regenerator.

2. In the heat exchanger, the wall temperature is constant in time, while in the regenerator it changes over a cycle in the form of a hysteresis loop.

The hysteresis loop reduces the heat flow transferred from the hot coolant to the cold one, since $\overline{\Delta t}_f$ the temperature difference between the coolants decreases by an amount commensurate with the width of the hysteresis loop. However, in thermal calculations of regenerators, the influence of the hysteresis loop on heat transfer is usually taken into account not through $\overline{\Delta t}_f$, but through

a special value called the non-stationarity coefficient. As a result, the actual heat transfer coefficient in the regenerator is determined by the formula

$$k = \varphi\, k_{\text{èä}} \qquad (1.1.5)$$

Here, $k_{\text{ид}}$ - is the non-stationarity coefficient; is the heat transfer coefficient of an "ideal" regenerator [5, 6, 8] related to the duration of a cycle consisting of two periods

$$k_{\text{ид}} = \left[\frac{1}{\alpha_0 \chi_0} + \frac{1}{\alpha_1 \chi_1} \right]^{-1} \qquad (1.1.6)$$

$\chi_j = T / T_{\text{ц}}$ – relative duration of the j-th period; T_j is the duration of the j-th period, c; $T_{\text{ц}} = \sum_{j=0}^{N_n - 1} T_j$ is the duration of the cycle, c; N_n is the number of periods in a cycle.

The value k_{id}, as follows from relation (2.1.6), depends only on the heat transfer coefficients and the relative duration of the periods. The coefficient φ takes into account the features of heat transfer associated with the non-stationary nature and inhomogeneity of the temperature fields in the packing. The value of the coefficient is strongly influenced by the frequency change of periods - cyclic frequency. With a decrease in the cyclic frequency, the width of the hysteresis loop increases and the no stationarity coefficient decreases. An increase in the cyclic frequency, on the contrary, leads to a narrowing of the loop and to an increase in φ and k. When the hysteresis loop degenerates into a straight line, then these quantities take on the values $\varphi = 1$ and $k = k_{\text{ид}}$. It follows that a regenerator in which the temperature of the packing surface changes linearly over a cycle can be called "ideal".

S. S. Kutateladze [6] proposed to consider an "ideal" regenerator in which the cyclic frequency $n \to \infty$. V.K. Migai [8] called a "ideal" regenerator in which the average temperature of the packing surface during the heating period is equal to that during the cooling period. How close are these definitions to the one formulated above?

At $n \to \infty$, the packing temperature changes linearly in time, and with a linear change in the packing surface temperature over a cycle, its average temperature over the cooling period will be equal to the average temperature over the heating period, so the definitions of the "ideal" regenerator given by Kutateladze and Migaem are essentially equivalent to the above .

It should be noted that with a linear change in time of the surface temperature, the temperatures of the internal points of the packing of the considered cross section will also change. In other words, in an "ideal" regenerator, the temperature fields in the direction of the normal to the packing surface are uniform. However, the converse statement of N.E. Ninua [1], that with a uniform nature of the temperature fields in the packing, the regenerator is “ideal”, is true only at a sufficiently high cyclic frequency.

The heat transfer coefficient k in the regenerator with an unsuccessful choice of the cyclic frequency may be insufficient even at high values of the heat transfer coefficients α_0 and α_1. Hence follows the conclusion about the need to improve the accuracy of determining the coefficient of non-stationarity in the thermal calculation of the regenerator.

1.1.2. Nozzle Temperature Estimation

As already noted, the vapors of certain substances contained in hot gases condense on the surface of the packing when its temperature drops below the dew point of these substances. Condensation and deposition of such substances on the surface of the packing create problems during the operation of the regenerator and the entire system in which the regenerator operates. So, for example, in the regenerator of an air-cooling machine, the condensation of water vapor in the form of frost can lead to clogging of the channels and a drop in the cooling capacity of the machine. To restore cooling capacity, one has to either switch to a standby regenerator or stop the machine.

As a preventive measure against clogging of the regenerator with frost, it is possible to recommend the installation of a separator-drop catcher in the

section where condensation begins. However, to correctly determine the location of its installation, it is necessary to know the temperatures of the packing in all sections of the regenerator, the number of which can reach several tens or even hundreds [9, 11].

In the regenerators of boilers operating on sulphurous fuel (fuel oil, solid fuel), first of all, sulfuric acid vapors contained in flue gases can condense [8]. The deposition of sulfuric acid drops on the surface of the packing leads to corrosion of the steel packing, a decrease in the heat transfer capacity of the regenerator, a decrease in the efficiency of the regenerator and the boiler unit, and subsequent failure of the regenerator [13].

A preventive measure against sulfuric acid corrosion in the regenerators of power steam generators is the preheating of air in the heater [13, 14] to a temperature of $t_0^{in} = 70 \ldots 76$ °C using steam taken from the turbine [10] used at thermal power plants (TPPs). However, this measure does not completely eliminate corrosion in the entire regenerator [10, 15]. Therefore, the nozzle along the path of heat carriers is divided into two parts - hot and cold. The hot part is located on the side of the inlet of hot gases, the cold part is located on the side of the inlet of cold air. The height of the hot part is selected so as to exclude sulfuric acid corrosion there. The possibility of corrosion in the cold part is not completely excluded, therefore, to increase the service life of the regenerator, the nozzle in the cold part is made of sheets of twice the thickness (1.2 mm) compared to sheets in the hot part (0.63 mm) [2, 8, 10].

From an energy point of view, air preheating is the cause of turbine power losses due to steam extraction to the heater and increased heat losses with flue gases. Calculations [16] show that a decrease in the heating temperature t_0^{in} from 76 to 55°C at an ambient temperature of 20°C makes it possible to reduce and respectively by 3.06 and 3.42 MW per one PP-950-250 steam generator. Therefore, it is very important to choose a value t_0^{in} that would allow, on the one

hand, to prevent sulfuric acid corrosion, and on the other hand, to ensure a minimum of losses ΔN_T and Q_2.

The reliability of the choice is due to the accuracy of determining the temperature of the nozzle. The existing methods of thermal calculation of the regenerator give only an approximate estimate. So, according to the normative method of thermal calculation of boiler units [10], it is recommended to estimate the minimum temperature of the packing of the i-th section by the formula

$$t_{w,i}^{\min} = \frac{\chi_1 \cdot \alpha_1 \cdot t_1'' + \chi_0 \cdot \alpha_0 \cdot t_0'}{\chi_0 \cdot \alpha_0 + \chi_1 \cdot \alpha_1} \qquad (1.1.7)$$

Where t_1'' and t_0' are the temperatures of the gas at the outlet of the i-th section and air at the inlet of the i-th section, K.

Formula (1.1.7) does not take into account the influence of such factors as cyclic frequency, type of material, thickness and length (height) of the leaf nozzle and therefore characterized by low reliability. This is confirmed by the recommendation contained in the normative method for calculating boiler units [10]: air before being supplied to the regenerator should be heated in a heater to such a temperature t_0^{in} at which the value obtained by formula (1.1.7) would be higher than the dew point by 10 - 15 K.

In regenerators of high-temperature installations (gas turbine engines, metallurgical furnaces, etc.), the accuracy of estimating the maximum temperature of the packing is important, which should not exceed the maximum allowable temperature for the material used. It is obvious that the estimate of this temperature using a formula similar to (1.1.7) will also have a large error.

§ 1.2. Review and analysis of RWP mathematical models

The first works devoted to the thermal calculation of the regenerator appeared in the 1920s. At the initial stage of the development of the theory of the regenerator, the calculation methods developed for the recuperator were transferred to the regenerator by refining the expression for the heat transfer coefficient. It was assumed that the packing temperature distribution along the regenerator is the same as along the recuperator. The search for refined expressions was carried out using approximate relations for the temperature distribution in the transverse section of the wall. This direction includes the works of V. Heiligenstedt [17], K. Rummel [18], A. Shack [12], in which methods for thermal calculation of regenerators with a brick packing for metallurgical furnaces were developed. The limited application of the methods proposed by them showed the significance of the influence on the heat transfer capacity of the regenerator of processes that are fundamentally different from the processes in the recuperator.

Subsequently, the theory of thermal processes in the regenerator followed the path of studying deviations from the temperature distribution inherent in the recuperator, focusing on temperature changes over time and neglecting the inhomogeneity of temperature fields across the packing. This direction was focused on regenerators with a high heat-conducting metal packing used in cryogenic technology and power engineering. The first works in this direction are devoted to the calculation of the initial heating or cooling at the start of the regenerator. These include the works of A. Anzelius [19], T. Schumann [20], A. Lovan [21]. In the works of V. Nusselt [22], H. Hausen [11], W. Schmeidler [23], G. Ackerman [24], S. S. Kutateladze [6], V. M. Datskovsky [5], V. TO. Blinking and collaborators [8], V.V. Faleev and A.V. Barakov [25] and others considered the steady state operation of the regenerator.

The theory of thermal calculation of the regenerator, taking into account the change in the temperature of the packing both in time and along the normal

to the wall, was developed by W. Nusselt, H. Hausen, V.N. Timofeev and colleagues. [26] and others.

A special place in the development of the analytical theory of the regenerator is occupied by the works of H. Hausen. He, independently of W. Nusselt, was the first to formulate the condition for switching periods in the regenerator, according to which the temperature fields in the packing at the end of one period are the initial ones for the next period, and thus described the processes in the regenerator as processes of temperature fluctuations. The formula he derived for the non-stationarity coefficient is used in engineering practice to the present day.

Most of the work relates to numerical methods for constructing mathematical models of the regenerator. These are the works of I.M. Gaisinsky [27], Yu.V. Matveev and V.K. Schukin [28], S.K. Popov [29], McDonald [30], Yu.D. Petrov [31], B. Bucklich [32, 33], A. Burns [34], A. Wilmot [35-39], J. Coppage and A. London [40], T. Lambertson [41], A. Lovan [42], M. Modest and S. Tien [43], J. Mondt [44], P. Racelos and M. Benjamin [45]. The listed works are devoted to obtaining expressions for the thermal efficiency of the regenerator and do not contain formulas for calculating the temperatures of the packing and coolants, as well as the non-stationarity coefficient, or they are given in such a form, which is very difficult to use. Indicative in this regard is the work of A. Wilmot and R. Dagan [46], in which the results of numerical calculations are generalized in the form of a series

$$\varphi = \frac{2}{\Omega}\left(\sum_{j=0}^{n}\frac{a_j \Lambda^{j+1}}{j+1} - \Lambda\right),$$

where Λ and Ω are the reduced length and time; a_j are the coefficients of the equation

$$\sum_{j=0}^{n} a_j (\Lambda - \xi)^j + e^{-\Omega}\sum_{j=0}^{n} a_j \xi^j + \sum_{j=0}^{n} a_j \int_0^{\xi} \varepsilon^j \cdot K(\xi - \varepsilon) d\varepsilon = 1$$

$\xi = \Lambda \frac{y}{L}$ is the dimensionless longitudinal coordinate.

Therefore, the consideration here of works on the numerical simulation of the regenerator is limited mainly to the staging part. It should also be noted that, despite the effectiveness of numerical methods, in most of the listed works, except for [27, 29], one-dimensional regenerator models are considered. In [27, 29], two-dimensional models with a spherical nozzle were constructed. The non-linear analogy method for calculating the characteristics of the regenerator is considered in [47, 48].

Simplifying Assumptions in the Regenerator Theory. The complexity of thermal processes in the regenerator does not allow describing all their features in full. Therefore, when developing the regenerator theory, the authors are forced to accept a number of simplifying assumptions and assumptions. Assumptions can be divided into generally accepted and special.

The commonly accepted assumptions are:

- the thermophysical properties of the packing and heat carriers, as well as the velocities of the latter along the flow, remain constant;
- heat transfer coefficients are constant for all parts of the packing surface;
- thermal conductivity of heat carriers in the direction of flow is negligible compared to the energy transfer by convection;
- heat carriers are not mixed during switching periods;
- the time of passage of coolant particles through the regenerator is negligible compared to the duration of the period;
- temperatures of heat carriers at the inlet to the regenerator are constant in time;
- transitions from one working period (cooling or heating) to another occur instantly, that is, the cycle does not contain "idle" periods;
- the ends of the sheet or cylindrical nozzle do not participate in heat exchange with heat carriers.

Special assumptions relate to the development of temperature fields in the packing and heat carriers, to the thermophysical properties of the packing, to the duration of heating and cooling periods, etc. The nature of these assumptions

largely depends on the type and purpose of the regenerator, on the type of packing. W. Nusselt proposed the following classification of special assumptions:

1) the time of one period is infinitesimal; the regenerator switches at an infinitely high frequency;

2) the thermal conductivity of the packing in all directions is infinitely large;

3) the thermal conductivity of the packing is zero in the direction of the flow and infinitely large along the normal to it;

4) the thermal conductivity of the packing is infinitely great in the direction of the flow and finite along the normal to it;

5) the thermal conductivity of the packing is zero in the direction of the flow and is finite along the normal to it.

The first version of this classification refers to the limiting mode of operation of the regenerator. The consideration of such a regime allowed H. Hausen [11] and S.S. Kutateladze [6] to obtain formulas for the heat transfer coefficient of an "ideal" regenerator.

The remaining assumptions are used in the models discussed below.

1.2.1. Regenerator models with infinitely high thermal conductivity of the packing in all directions

The assumption of a high thermal conductivity of the packing made it possible to consider the temperature fields in it to be homogeneous and dependent only on time. This assumption is accepted in the works of H. Hausen [11], S.S. Kutateladze [6], V.M. Datskovsky [5], V.K. Blinking [8].

At a high frequency of switching periods, the temperature waves do not have time to noticeably penetrate into the wall thickness and, therefore, the temperature of the packing remains practically unchanged. Under these conditions, there is no need to determine the temperature fields in the packing - it is enough to consider the heat exchange of the packing surface with heat carriers. As a result of this consideration, H. Hausen obtained a formula for the heat transfer coefficient of an "ideal" regenerator with the same period durations:

$$k_{id} = \left(\frac{1}{\alpha_0} + \frac{1}{\alpha_1} \right)^{-1}, \qquad (1.2.1)$$

where the heat transfer coefficient k_{id} is related to the duration of one period.

S.S. Kutateladze, in addition to the assumption of a high thermal conductivity of the packing, makes another assumption - that the temperatures of the heat carriers are constant both along the flow $t_{f,0}$ and $t_{f,1}$ in time, which implies

From the heat balance of the packing:

$$c_w M_w dt_{w,j} = \alpha_j F_{w,j} (t_{w,j} - t_{f,j}) d\tau$$

a differential equation was derived, which in dimensionless variables can be written as follows:

$$\frac{d\theta_{w,j}}{dF_0} = -B\mathrm{i}_j (\theta_{w,j} - \theta_{f,j}) \qquad (1.2.2)$$

Here, $\theta_{w,j} = 2(t_{w,j} - t_{f,\min})/(t_{f,\max} - t_{f,\min})$ is the relative packing temperature; $\theta_{w,j} = 2(t_{f,j} - t_{f,\min})/(t_{f,\max} - t_{f,\min})$ is relative temperature of the coolant at $(\theta_{f,0} = 0,\ \theta_{f,1} = 2)$; $t_{f,\max}$ is the smallest value of the coolant

temperatures, K; $t_{f,\max}$ is the highest value of the coolant temperatures, K; $F_0 = 4a_w\tau/\delta_w^2$ is the current Fourier number ($0 \le F_0 \le F_{0T_j}$); τ is time from the beginning of the period, s; $F_{0T_j} = 4a_wT_j/\delta_w^2$ is the limiting Fourier number of the j-th period; $a_w = \lambda_w/(\rho_w c_w)$ is coefficient of thermal diffusivity of the packing, m^2/s; λ_w is coefficient of thermal conductivity of the nozzle, W / (m-K); ρ_w is packing density, kg/m^3; c_w specific heat capacity of the packing, J/(kg K); δ_w is packing thickness, m; $Bi_j = \alpha_j\delta_w/(2\lambda_w)$ is the Bio number of the j-th period.

Under the initial condition $\theta_{w,j}(F_0 = 0) = \theta_{w,j}(0)$ and the condition of constancy $\theta_{w,j}$, the solution to equation (1.2.2) is the function:

$$\theta_{w,j} = \theta_{f,j} - [\theta_{f,j} - \theta_{w,j}(0)]\exp(-Bi_jF_0) \tag{1.2.3}$$

Using switching conditions:

$$\theta_{w,0}(0) = \theta_{w,1}(F_{0T_1}) \text{ and } \theta_{w,1}(0) = \theta_{w,0}(F_{0T_0}),$$

From formulas (1.2.3), the initial packing temperatures were determined:

$$\begin{aligned}\theta_{w,0}(0) &= \frac{\theta_{f,0}[1-\exp(-m_0)]\exp(-m_1) + \theta_{f,1}[1-\exp(-m_1)]}{1-\exp(-m_0-m_1)}\\ \theta_{w,1}(0) &= \frac{\theta_{f,0}[1-\exp(-m_0)] + \theta_{f,1}[1-\exp(-m_1)]\exp(-m_0)}{1-\exp(-m_0-m_1)}\end{aligned} \tag{1.2.4}$$

Where $m_j = \alpha_j\chi_jF_w/W_w = Bi_jF_{0T_j}$; $W_w = M_wc_wn$ is the water equivalent of the packing, W/K; M_w is the mass of the packing, kg; n is cyclic frequency, s^{-1}. After substituting formulas (1.2.3) and (1.2.4) into the equation for the amount of heat transferred by the nozzle per unit time from the hot coolant to the cold

$$Q = \alpha_0F_{w,0}(\bar{t}_{w,0} - t_{f,0}) = \alpha_1F_{w,1}(t_{f,1} - \bar{t}_{w,1}),$$

where $\bar{t}_{w,j}$ is the packing temperature averaged over the j-th period in K, the following equation was obtained for the regenerator heat load:

$$Q = 2c_w\rho_w\delta_wF_wn(t_{f,1} - t_{f,0})\frac{[1-\exp(-m_0)][1-\exp(-m_1)]}{1-\exp(-m_0-m_1)} \tag{1.2.5}$$

and the formula for the non-stationarity coefficient:

$$\varphi = \left(\frac{1}{m_0} + \frac{1}{m_1} \right) \frac{[1 - \exp(-m_0)][1 - \exp(-m_1)]}{1 - \exp(-m_0 - m_1)} \tag{1.2.6}$$

Assuming that the regenerator transmits the largest heat load at an infinitely high cycle switching frequency ($n \to \infty$, $F_{0T_j} \to 0$, $m_j \to 0$), Kutateladze, after the disclosure of uncertainty in (1.2.5), derived the formula:

$$Q_{\max} = \frac{t_{f,1} - t_{f,0}}{\frac{1}{\alpha_0 F_{w,0}} + \frac{1}{\alpha_1 F_{w,1}}} = F_w \frac{t_{f,1} - t_{f,0}}{\frac{1}{\alpha_0 \chi_0} + \frac{1}{\alpha_1 \chi_1}}$$

from the comparison of which with the similar one for the recuperator follows the formula (1.2.6) for the heat transfer coefficient of the "ideal" regenerator with different durations of periods.

Formula (1.2.6) gives underestimated values of the coefficients of non-stationarity and heat transfer, since the assumption of the invariance of the temperatures of the heat carriers in time and along the flows does not take into account the finiteness of the heat capacity of the heat carrier flows.

V.M. Datskovsky considered the heat transfer of heat carriers with a packing surface element of length *dy* and the change in temperature of the entire packing over a time interval $d\tau$. Using the method of elementary balances, differential equations for coolants and nozzles were derived:

$$\begin{aligned} \frac{\partial t_{f,j}}{\partial y} &= \frac{A_j}{l} \left[t_{w,j} - t_{f,j}(y) \right]; \\ \frac{\partial t_{w,j}}{\partial \tau} &= B_j \left[t_{f,j}(y=0) - t_{f,j}(y=l) \right], \end{aligned} \tag{1.2.7}$$

Where $A_j = \frac{\alpha_j F_w \chi_j}{W_{f,j}}$; $B_j = \frac{W_{f,j} n}{W_w \chi_j}$.

Assuming a high thermal conductivity of the packing, the value $t_{w,j}$ becomes dependent only on time. Integration of differential equations (1.2.7) using the boundary condition $t_{f,j}(y=0) = t_{f,j,0}$ for heat carriers gave expressions for the temperatures of heat carriers and nozzles:

$$t_{f,j} = t_{f,j,0} - \left[t_{f,j,0} - t_{w,j}(0)\right]\left[1 - \exp(-A_j y / l)\right]\exp(-M_j);$$
$$t_{w,j} = t_{f,j,0} - \left[t_{f,j,0} - t_{w,j}(0)\right]\exp(-M_j \tau), \qquad (1.2.8)$$

Where $t_{w,0}(0)$ and $t_{w,1}(0)$ are the packing temperatures at the beginning of the periods, are found from the switching conditions $t_{w,0}(0) = t_{w,1}(T_1)$ and $t_{w,1}(0) = t_{w,0}(T_0)$:

$$t_{w,0}(0) = \frac{t_{f,0,0}\left[1 - \exp(-M_0 T_0)\right]\exp(-M_1 T_1) + t_{f,1,0}\left[1 - \exp(-M_1 T_1)\right]}{1 - \exp(-M_0 T_0 - M_1 T_1)};$$

$$t_{w,0}(0) = \frac{t_{f,0,0}\left[1 - \exp(-M_0 T_0)\right] + t_{f,1,0}\left[1 - \exp(-M_1 T_1)\right]\exp(-M_0 T_{01})}{1 - \exp(-M_0 T_0 - M_1 T_1)};$$

The substitution of the derived expressions for temperatures into the heat balance equations of the regenerator allowed Datskovsky to obtain a formula for the nonstationarity coefficient (1.2.6), in which the quantities

$$M_j \frac{W_{f,j}}{M_w c_w}\left[1 - \exp(-A_j)\right]. \qquad (1.2.9)$$

Formula (1.2.6) together with (1.2.9) gives overestimated values of the non-stationarity coefficient, the reason for which is: 1) the assumption that the packing temperature is constant along the coolant flows and 2) the neglect of the local derivative of the coolant temperature with respect to time in the first equation of the system (1.2.7).

V.K. Migai, considering a rotating disk-type regenerator with a sheet metal packing, made up the thermal balances of the packing element and coolants using contradictory assumptions:

- thermal conductivity of the nozzle in all directions is infinitely large;
- the temperature of heat carriers along the flows varies according to a linear law.

Neglecting the convective component of the time derivative in their differential equation for the energy of the coolant, the authors of [8] obtained expressions for the temperatures of the nozzles when the rotor rotated with a

frequency n, through an angle w at constant temperatures of the coolants at the inlet to the regenerator:

$$t_{w,j} = t_{f,j}(0) \pm \Delta t_{f,\max} \frac{1-\exp(-b_j)}{1-\exp(b_0-b_1)} \exp(-b_j \tau / T_j). \qquad (1.2.10)$$

Here $t_{f,j}(0)$ is the temperature of the j-th coolant at the inlet to the regenerator, K; $\Delta t_{f,1\max} = t_{f,1}(0) - t_{f,0}(0)$ is maximum temperature difference between heat carriers; $b_j = \frac{1}{F_w}\left(\frac{1}{m_j} + \frac{W_w}{2W_{f,j}}\right)^{-1}$; $\tau = \frac{\omega}{2\pi n}$ is the time of rotation of the rotor at an angle w; time of rotation of the rotor by an angle w; the sign "+" refers to the cold period $(j=0)$, the sign "-" - to the hot $(j=1)$.

Average nozzle temperatures over the period:

$$\bar{t}_{w,j} = t_{f,j}(0) \pm \frac{\Delta t_{f,\max}}{b_j} \frac{[1-\exp(-b_0)][1-\exp(-b_1)]}{1-\exp(-b_0-b_1)}.$$

Average temperatures of heat carriers for the period:

$$\bar{t}_{f,j} = t_{f,j}(0) \pm \Delta t_{f,\max} \frac{W_w}{2W_{f,j}} \frac{[1-\exp(-b_0)][1-\exp(-b_1)]}{1-\exp(-b_0-b_1)}. \qquad (1.2.11)$$

Formula for the coefficient of non-stationarity:

$$\varphi = \frac{1/m_0 + 1/m_1}{\frac{1-\exp(-b_0-b_1)}{[1-\exp(-b_0)][1-\exp(-b_1)]} - \left(\frac{1}{W_{f,0}} + \frac{1}{W_{f,2}}\right)\frac{W_w}{2}}. \qquad (1.2.12)$$

1.2.2. Models of the regenerator with zero thermal conductivity of the packing in the direction of the coolant flow and infinitely large along the normal to the flow

This assumption implies a packing in the form of a layer of metal balls or granules, when the small contact area of the granules with each other prevents the transfer of energy by thermal conductivity from granule to granule. It was used by A. Anzelius [19], W. Nusselt [22], T. Schumann [20], J. Coppage and A. London [40], T. Lambertson [41], J.R. Mondtom [44], A.J. Wilmot [35-39], B.S. Bucklich [32, 33].

The solutions of Anzelius-Shuman, J. Coppage and A. London have found practical application.

Under the assumption under consideration, the first regenerators were analytically studied independently by A. Anzelius [19], W. Nusselt [22], T. Schumann [20].

A. Anzelius and later T. Schumann compiled two differential equations in partial derivatives as applied to a porous body through which a coolant is pressed.

$$\frac{\partial t_f}{\partial Z} = t_w - t_f, \quad \frac{\partial t_f}{\partial Y} = t_f - t_w. \tag{1.2.13}$$

Here $Y = \alpha F_w / c_p G_f$ - is the relative heat transfer; $Z = \alpha F_w \tau / c_w M_w$ - relative time. The initial body temperature was assumed to be constant.

Boundary and initial conditions of the system of equations (1.1.13):

$$\begin{aligned} &at \quad x = 0: \; t_f = t_{f,0} \\ &at \quad \tau = 0: \; t_w = t_{w,0}. \end{aligned} \tag{1.2.14}$$

Anzelius presented the solution of problem (1.2.13), (1.2.14) in integral form. Thanks to the assumption of a homogeneous initial temperature field, Schumann was able to give a solution in the form of infinite series

$$\frac{t_f}{\Delta t_f} = 1 - \exp(-Y - Z)\sum_{n=1}^{\infty} Y^n W_n = \exp(-Y - Z)\sum_{n=1}^{\infty} Z^n W_n;$$
$$\frac{t_w}{\Delta t_f} = 1 - \exp(-Y - Z)\sum_{n=1}^{\infty} Y^n W_n = \exp(-Y - Z)\sum_{n=1}^{\infty} Z^n W_n, \quad (1.2.15)$$

Where $W_n = \frac{d^n}{d(YZ)^n} I_0\left(2\sqrt{YZ}\right)$; Δt_f is the temperature head of the coolant at the nozzle inlet; I_0 - is the modified Bessel function of the first kind of order zero.

The assumption that the initial body temperature is constant makes it possible to apply solution (1.2.15) only to estimate the initial heating or cooling of the packing at the very beginning of the regenerator operation. Ignoring the convective part of the time derivative of the coolant temperature in the first equation of system (1.2.13) leads to an overestimation of the rate of change of this temperature over time.

V. Nusselt, bearing in mind the cyclic nature of the processes in the regenerator, considered the initial temperature for the period in the packing to be some function of the transverse coordinate. Using the method of heat balances for the element of the wall surface and assuming that $\partial t_f / \partial \tau = 0$ it is independent of Anzelius and Schumann, he derived equations for the temperatures of the packing and coolant:

$$\frac{\partial t_f}{\partial y} = -a(t_w - t_f), \quad \frac{\partial t_w}{\partial \tau} = b(t_f - t_w), \quad (1.2.16)$$

Where $a = \alpha B/(G_f c_p)$; $b = \alpha F_w/(M_w c_w)$; B is the channel width, m; G_f is the mass flow rate of the coolant, kg/s; c_p - isobaric heat capacity of the coolant, J / (kg-K).

Boundary and initial conditions of equations (1.2.16):

$$at \quad x = 0: \; t_f = t_{f,0}$$
$$at \quad \tau = 0: \; t_w = F(y) \quad (1.2.17)$$

Equations (1.2.16) Nusselt transformed into a differential equation

$$\frac{\partial^2 t_f}{\partial \tau \partial y} + a\frac{\partial t_f}{\partial \tau} + b\frac{\partial t_f}{\partial y} = 0,$$

whose integration gave the expression for the coolant temperature

$$t_f = e^{-ay-b\tau}\left\{t_{f,0} I_0\left(2\sqrt{aby\tau}\right) + a\int_0^y e^{a\xi} F(\xi) I_0\left[2\sqrt{aby(y-\xi)}\right]d\xi + \right.$$
$$\left. + bt_{f,0}\int_0^\tau e^{b\eta} I_0\left[2\sqrt{aby(\tau-\eta)}\right]d\eta\right\}, \tag{1.2.18}$$

and then the wall temperature

$$t_w = e^{b\tau} F(y) e^{-ay-b\tau}\left\{t_{f,0}\sqrt{\frac{b\tau}{ay}} I_1\left(2\sqrt{aby\tau}\right) + \right.$$
$$+ \int_0^y e^{a\xi} F(\xi)\sqrt{\frac{ab\tau}{y-\xi}} I_1\left[2\sqrt{aby(y-\xi)}\right]d\xi + \tag{1.2.19}$$
$$\left. + \frac{bt_{f,0}}{a}\int_0^\tau e^{b\eta}\sqrt{\frac{ab(\tau-\eta)}{y}} I_1\left[2\sqrt{aby(\tau-\eta)}\right]d\eta\right\},$$

here I_1 - is a modified Bessel function of the first kind of the first order.

Unfortunately, Nusselt could not give a complete solution to equations (1.2.15) and (1.2.16), since the question of the temperature field in the nozzle at the beginning of the period, that is, the function, remained open.

NOTE. Ninua [1], assuming the temperature fields in the packing to be homogeneous, considered the regenerator to be “ideal”, taking the no stationarity coefficient equal to unity.

In the work of J. Coppage and A. London [40], the following equations are derived from the heat balance of the packing element:

$$G_{f,j} c_{p,j}\frac{\partial t_{f,j}}{\partial y} + \rho_{f,j} F_{k,j} c_{p,j}\frac{\partial t_{f,j}}{\partial \tau} = \pm\frac{M_w c_w}{L}\frac{\partial t_w}{\partial \tau},$$
$$G_{f,j} c_{p,j}\frac{\partial t_{f,j}}{\partial y} + \rho_{f,j} F_{k,j} c_{p,j}\frac{\partial t_{f,j}}{\partial \tau} = \pm\frac{\alpha_w F_w}{L}(t_{f,j} - t_w) \tag{1.2.20}$$

which are solved under boundary conditions

$$t_{f,j}(y=0,\ \tau) = const. \tag{1.2.21}$$

In equations (1.2.20), «+» refers to the cold period; «–» - to hot; $F_{k,j}$ - is the cross-sectional area of the channel in the j-th period, m^2.

The results of the numerical solution of the system (1.2.20) and (1.2.21) were later generalized and presented as a formula for the thermal efficiency of the regenerator

$$\eta_t = \frac{1-\exp\left[-NTU\left(1-W_{f,\min}/W_{f,\max}\right)\right]}{1-\exp\left[-NTU\left(1-W_{f,\min}/W_{f,\max}\right)\right]W_{f,\min}/W_{f,\max}} \times \left[1-\frac{1}{9}\left(\frac{W_{f,\min}}{W_w}\right)^2\right]. \quad (1.2.22)$$

Here $\eta_t = \dfrac{W_{f,0}(t''_{f,0}-t'_{f,0})}{W_{f,\min}(t'_{f,1}-t'_{f,0})} = \dfrac{W_{f,1}(t'_{f,1}-t''_{f,1})}{W_{f,\min}(t'_{f,1}-t'_{f,0})}$ is the thermal efficiency; $NTU = k_{id}F_w/W_{f,\min}$ is the number of transfer units.

With a known value of the quantity, the nonstationarity coefficient is then determined by the formulas:

$$\phi = \frac{Q}{k_{id}F_w\overline{\Delta t_f}}; \quad Q = W_{f,\min}\eta_t(t'_{f,1}-t'_{f,0})\frac{(1/W_{f,1}-1/W_{f,0})}{\ln\left(\dfrac{1-\eta_t W_{f,\min}/W_{f,0}}{1-\eta_t W_{f,\min}/W1}\right)} \quad (1.2.23)$$

T. Lambertson [41] gives a numerical solution of the heat balance equations for the elements of a rotating axial rotor (disk-type regenerator). Also, a numerical solution for a rotating regenerator of a radial (drum) type is considered by J. Mondt [44]. In [43], a thermodynamic analysis of the regenerator is given by the perturbation method. A. Wilmot et al: [35-39] build regression mathematical models in the form of a series based on the solution of differential equations for the one-dimensional model of W. Nusselt (1.2.13), (1.2.14). The target function is the thermal efficiency of the regenerator, the factors are the reduced length and reduced time proposed by H. Hausen [11] (see below). A similar approach to regenerator modeling was used in the works of B. Baklich et al. [32, 33]. An attempt to take into account the influence of the variability of the mass flow rates of coolants on the thermal efficiency of the regenerator was made in [45].

1.2.3. Models of a regenerator with an infinitely high thermal conductivity of the packing in the direction of the coolant flow and a final thermal conductivity along the normal to the flow

The assumption of an infinitely large thermal conductivity of the packing in the direction of flow assumes that the temperature of the packing in this direction is constant. This assumption was adopted in the first works on the creation of the theory of the regenerator - in the works of W. Heiligenstedt [17], K. Rummel [18] and W. Nusselt [22] and therefore are mainly of historical interest.

V. Heiligenstedt, assuming that the temperatures of the heat carriers remain constant along the path and in time, and the temperature of the packing changes only in the transverse direction, used a particular solution of the one-dimensional non-stationary heat conduction equation for a flat wall with a thickness δ_w:

$$t_w = t_f + C\exp(\beta^2 a_w \tau)\cos\beta\left(x - \frac{\delta_w}{2}\right), \qquad (1.2.24)$$

Where C - is a constant; β - is the first root of the characteristic equation:

$$\beta \cdot tg\left(\frac{\beta\delta_w}{2}\right) = \frac{\alpha}{\lambda_w}.$$

The calculation of the thermal state of the packing according to Heiligenstedt begins with a uniform temperature distribution in the cross section and continues until a steady state is reached. The constant C for each period is calculated from the temperature distribution at the end of the previous period.
Obviously, the modeling of thermal processes in the regenerator by equation (1.2.24) is very approximate. The most significant disadvantages of this model:

1) In reality, the solution of the differential equation of heat conduction is an infinite series in terms of the values of the roots of the characteristic equation. The replacement of a series by its first member can be used with a known error only for very small values of the Biot number ($Bi < 10^{-4}$). The nozzles of a large thickness δ_w of low thermally conductive material, as in the

case of brickwork, which Heiligenstedt had in mind, are characterized by values of Bi> 1.

2) One of the features of thermal processes in the regenerator packing is that the temperature field at the end of one period is the initial one for the next period (switching condition). Equation (1.2.24) cannot satisfy this condition at all points of the nozzle cross section. Therefore, Heiligenstedt selects the values of the constant C so that this condition is satisfied for the temperature averaged over the wall thickness. As a result, the transverse temperature profile, bent downwards at the end of the heating period, instantly bends at the moment of switching periods and turns into an upwardly bent profile with the same average temperature. Such a development of temperature fields in the packing is contrary to common sense.

The merit of W. Heiligenstedt is that he was the first to propose a formula for the heat transfer coefficient of the regenerator:

$$\mathrm{k} = \frac{\rho_w c_w \delta_w}{2(T_1 + T_2)} \frac{(1-\psi_1)(1-\psi_2)}{1-\psi_1\psi_2},$$

Where $\psi_1 = \exp(-\beta_1^2 a_w T_1)$; $\psi_2 = \exp(-\beta_2^2 a_w T_2)$; β_1, β_2 are the roots of the characteristic equation for the periods of cooling and heating.

K. Rummel, assuming that the amplitude of the temperature of the packing surface per cycle is proportional to the amplitude of the temperatures averaged over the period of the same packing, that is

$$t_{w,\max} - t_{w,\min} = \varsigma(\bar{t}_{w,\max} - \bar{t}_{w,\min}),$$

proposed an empirical method for determining the heat transfer coefficient by a formula that is similar in appearance to the expression for the heat transfer coefficient of a flat wall of a recuperative heat exchanger:

$$\frac{1}{k} = (T_1 + T_2) = \left(\frac{1}{\alpha_1 T_1} + \frac{1}{\alpha_2 T_2} + \frac{2}{\varsigma\eta\rho_w c_w \delta_w} \right), \qquad (1.2.25)$$

Where η - is the degree of heat utilization by the packing, which is the ratio of the heat accumulated by the packing to the heat that could be accumulated with an infinitely large thermal conductivity of the packing; ς - coefficient of proportionality.

Empirically, Rummel found that for blast-furnace air heaters, ς the coefficient usually varies within 2 - 3.5. He proposed to calculate the function by the approximate formula

$$\frac{1}{\eta} = 1 + \frac{\delta_w^2}{4a_w(T_1 + T_2)}.$$

The empirical nature of formula (1.2.25) η and ζ the functions included in it limited the scope of its application to air heaters for blast furnaces (coopers).

To determine the temperature distribution in the direction transverse to the surface of the body, W. Nusselt wrote down the boundary value problem of heat conduction in a one-dimensional setting, including the differential equation of heat conduction in the plate:

$$\frac{\partial t_w}{\partial \tau} = a_w \frac{\partial^2 t_w}{\partial x^2}, \tag{1.2.26}$$

initial condition:

$$t_w(x,0) = f(x) \tag{1.2.27}$$

and boundary conditions, one of which is the condition of symmetry of temperature fields in the middle plane of the plate ($x = \delta_w$):

$$\frac{\partial t_w(\delta_w, \tau)}{\partial x} = 0, \tag{1.2.28}$$

and others - the heat transfer equation of the surface ($x = 0$) of the body:

$$\alpha\left[t_f(y,\tau) - t_w(0,\tau)\right] = \lambda_w \frac{\partial t_w(0,\tau)}{\partial x}. \tag{1.2.29}$$

Here x is the transverse coordinate, m. To determine the heat carrier temperature included in equation (1.2.29), $t_f(y,\tau)$ W. Nusselt writes the flow energy equation:

$$\alpha\left(t_f - t_w\right) = -\frac{G_f c_p}{b}\left(\frac{\partial t_f}{\partial y} + \frac{1}{w_f}\frac{\partial t_f}{\partial \tau}\right),$$

Where b is the width of the channel, m.

Having accepted the assumption $1/w_f << 1$ and considering the value to be $\frac{1}{w_f}\frac{\partial t_f}{\partial \tau}$ infinitely small, i.e. neglecting the energy costs for heating (cooling) the coolant when its temperature changes over time, Nusselt obtained a simplified equation for the coolant in the form

$$\alpha\left(t_f - t_w\right) = -\frac{G_f c_p}{b}\frac{\partial t_f}{\partial y}, \qquad (1.2.30)$$

and the boundary condition for it

$$t_f(y=0) = t_{f,0} \qquad (1.2.31)$$

The solution of the boundary value problem (1.2.26)-(1.2.29) according to Nusselt is the series:

$$t_w = t_{f,0} + \sum_{n=0}^{\infty} N_n \exp(-\gamma_n^2 a_w \tau)\sin(\gamma_n x + \beta_n),$$

Where γ_n and β_w are the roots of the characteristic equations; N_n - coefficients:

$$N_n = 2\frac{t_{f,0}\cdot\cos\beta_n + \gamma_n\int_0^{\delta_w} f(x)\cdot\sin(\gamma_n x + \beta_n)dx}{\gamma_n\delta_w + \sin(2\beta_n)/2}.$$

The coolant temperature is a solution to problem (1.2.30), (1.2.31):

$$t_f = t_w + (t_{f,0} - t_w)\cdot\exp\left(-\frac{\alpha b}{G_f c_p}y\right).$$

As in the problem considered in Section 1.2.3, here Nusselt also does not give a complete solution, since the question of the temperature field in the

packing at the beginning of the period, that is, of the function f(x), continued to be open.

Nusselt's merit is that he was the first to draw attention to the inhomogeneity of the temperature fields in the packing at the beginning of each period. In addition, independently of H. Hausen, he combined the boundary value problem of heat conduction in a wall and the problem of energy transfer by a coolant into one problem - the conjugate problem of cyclic heat transfer of a solid body with several media.

1.2.4. Models of the regenerator with zero thermal conductivity of the packing in the direction of the coolant flow and the final one along the normal to the flow

As a packing, a bulk material with low thermal conductivity is assumed here, for example, rocks, ceramics, brickwork, etc. This assumption was used by W. Nusselt [22], H. Hausen [11], A. Shack [12], V.N. Timofeev [26], A. Lovan [42]. W. Nusselt models this approximation with a system of equations (1.2.26) -(1.2.29), replacing the one-dimensional initial condition (1.2.26) with a two-dimensional one:

$$t_w(x, y, 0) = \Phi(x, y) \qquad (1.2.32)$$

An analysis of the Nusselt model shows that it is internally contradictory: according to condition (1.2.32), the temperature fields in the packing must be two-dimensional, and the differential heat conduction equation (1.2.26) provides for temperature changes in only one coordinate - the transverse one. The author does not give a complete analytical solution of this problem.

X. Hausen considers a regenerator consisting of a packing in the form of flat plates of the same thickness δ_w, between which gas is blown at a uniform and equally directed speed.

The process of heat transfers from a hot coolant to a packing and from a packing to a cold coolant was described by X. Hausen, independently of W. Nusselt, as an adjoin problem, in which he included the boundary value problem of heat conduction in the packing and the problem of heat transfer by coolants.

In accordance with the accepted assumptions, he considered the boundary value problem of one-dimensional heat conduction in a packing, similar to problem (1.2.26)-(1.2.29), with one difference - instead of the non-specific initial condition (1.2.27) X. Hausen proposed the switching condition. According to this condition, the temperature at each point of the packing at the beginning of the period is equal to the corresponding temperature at the end of the previous period. Therefore, the boundary value problem of heat conduction according to X. Hausen took the form of a system containing the differential heat conduction equation (1.2.26), boundary conditions (1.2.28) and (1.2.29) and the switching condition:

$$t_{w,1} = t_{w,2}. \tag{1.2.33}$$

In addition, Hausen also uses an equation of the form (1.2.2).

Neglecting, like Nusselt, the energy costs for heating (cooling) heat carriers, which are a consequence of nonstationarity, Hausen derived a differential equation for convective energy transfer by heat carriers (1.2.30).

The joint solution of the problem consisting of equations (1.2.26), (1.2.28)-(1.2.30), (1.2.32), Hausen carried out under the assumption that the longitudinal temperature gradients of the packing and coolants are the same, which is equivalent to the condition about the constancy of temperature differences between the heat carriers and the nozzle along the entire length of the latter:

$$\frac{\partial t_w}{\partial y} = \frac{\partial \overline{t_w}}{\partial y} = \frac{\partial t_f}{\partial y}. \tag{1.2.34}$$

The solution to the boundary value problem (1.2.26), (1.2.27) - (1.2.30), (1.2.32) is the expression:

$$t_w = A + \frac{\partial \overline{t_w}}{\partial \tau} \cdot \tau - \frac{1}{2a_w} \frac{\partial \overline{t_w}}{\partial \tau} \cdot x \cdot (\delta_w - x) + \\ + \sum_{i=0}^{\infty} B_i \cdot \exp(-\beta_i^2 a_w \tau) \cdot \cos \beta_i \left(x - \frac{\delta_w}{2} \right), \tag{1.2.35}$$

where A, B_i and β_i are constants.

The condition (1.2.34) adopted by Hausen makes the problem contradictory, since $t_f - t_{w,0} = const$ at from equation (1.2.2) it follows that the temperature of the packing, averaged over the section, depends linearly on time:

$$\frac{\partial \overline{t_w}}{\partial \tau} = const \tag{1.2.36}$$

To fulfill condition (1.2.36), Hausen selects the constants A and β_i in equation (1.2.35) so that the values of the terms of the series after integration (1.2.35) over the packing thickness vanish. This implies:

$$A = \overline{t_w}, \quad \beta_i = \frac{2\pi i}{\delta_w}. \tag{1.2.37}$$

The values of the β_i constants according to the formula (1.2.37) contradict the characteristic equation of the heat conduction problem in the packing:

$$\beta \cdot tg(\beta \delta_w / 2) = \frac{\alpha}{\lambda_w}, \tag{1.2.38}$$

according to which $\beta_i = 2\pi i / \delta_w$ the values correspond to a case of no practical interest $\alpha = 0,$. Since in reality the values of the roots β_i are nevertheless determined by the equation (1.2.38), then the assumption (1.2.34) is invalid.

The Hausen B_i constants are found from the switching condition (1.2.33). After determining by equation (1.2.35) at the value of $x = 0$ the nozzle surface $t_{w,0}$ temperature t_f, the coolant temperature is found by $t_{w,0}$ adding to the time-invariant difference $t_f - t_{w,0}$.

The results of calculations by the numerical-graphical method of the temperatures of the packing and heat carriers, together with the heat balance

equation of the regenerator, were generalized in the form of an equation for the heat transfer coefficient of the regenerator [1]:

$$\frac{k}{k_0}=\left\{1-\frac{1}{\Lambda}\left[0{,}8\Omega-3\mathrm{th}(\Omega/5)\right]\right\}\left[1+\frac{\frac{1}{\chi_1}+\frac{1}{\chi_2}}{\frac{1}{\alpha_1\chi_1}+\frac{1}{\alpha_2\chi_2}}\frac{\delta_w}{\lambda_w}\Phi\right]^1,$$

Where $\Lambda=4k_0F_w\frac{T_0+T_1}{W_{f,0}T_0+W_{f,1}T_1}$ is the reduced length; $\Omega=2k_0F_w\frac{T_0+T_1}{W_w}$ - reduced time; k_0 is the heat transfer coefficient due to the fundamental harmonic of the packing temperature fluctuation; Φ is a function that reflects the effect of rapid changes in the temperature of the surface of the packing and coolants immediately after switching periods.

The values of k_0 and Φ Hausen proposed to calculate according to the formulas:

$$\frac{1}{k_0}=\frac{1}{\alpha_0\chi_0}+\frac{1}{\alpha_1\chi_1}\left(\frac{1}{\chi_0}+\frac{1}{\chi_1}\right)\frac{\delta_w}{\lambda_w}\Phi;$$

$$\Phi=\begin{cases}\frac{1}{6}-0{,}00556\frac{\delta_w^2}{2a_w}\left(\frac{1}{T_0}+\frac{1}{T_1}\right) & at\ \ \frac{\delta_w^2}{2a_w}\left(\frac{1}{T_0}+\frac{1}{T_1}\right)\le 10;\\ 0{,}357\sqrt{0{,}3+\frac{\delta_w^2}{2a_w}\left(\frac{1}{T_0}+\frac{1}{T_1}\right)} & at\ \ \frac{\delta_w^2}{2a_w}\left(\frac{1}{T_0}+\frac{1}{T_1}\right)>10.\end{cases}$$

In industrial regenerators, usually $\Omega/\Lambda<0{,}5$. In such cases $\Phi\to 0,\ k_0\to k_{id}$ and from the expression for the formula known in engineering practice for the k/k_0 nonstationary coefficient is obtained [9, 11]:

$$\phi=1-\frac{0{,}8\Omega-3\mathrm{th}(\Omega/5)}{\Lambda}\ at\ \ \Omega/5<0{,}5 \qquad (1.2.39)$$

The approximate method for calculating A. Shak's regenerators is designed for air heaters of metallurgical furnaces with a packing of refractory bricks and is based on the use of the empirical formulas proposed by him for the temperatures of the packing (brick) and gas:

$$t_w(0) - \overline{t_w} = C_1 + C_2 \exp\left(\frac{\varsigma_1 \tau}{T}\right); \quad t_f = C_3 + C_4 \exp\left(\frac{\varsigma_2 \tau}{T}\right),$$

Where C_1, C_2,…,C_4, ς_1 and ς_2 are constants $t_w(0)$ and $\overline{t_w}$ are the surface temperature and the packing temperature averaged over the period, K.

Shack determined the values by C_1, C_2,…,C_4 an approximate analysis of the fluctuations in time of gas temperatures during periods of heating and cooling, using the ratio for the degree of heat utilization η and the switching condition. From a comparison with experimental data, A. Shak found that $\varsigma_1 = -8$ and $\varsigma_2 = 0{,}1$. Substituting the obtained expressions for $t_{w,0} - \overline{t_w}$ and t_f into the differential equation derived from the heat balance of the packing element

$$\frac{\partial \overline{t_w}}{\partial \tau} = \frac{2\alpha}{\rho_w c_w \delta_w}[t_f - t_w(0)],$$

A. Shak integrated it relatively and, after intermediate transformations, obtained an expression for the coefficient of non-stationarity [12]

B.

$$\varphi = U / V, \tag{1.2.40}$$

Where

$$U = \frac{1-\exp(-m)}{m} + 9{,}52\frac{\Delta t_{f,0} + \Delta t_{f,1}}{\bar{t}_{f,0} - \bar{t}_{f,1}} \times$$
$$\times \left\{1{,}05\frac{1-\exp(-m)}{m} - \left[1{,}05 - \frac{1-\exp(-m)}{m}\right]\frac{m}{0{,}1+m}\right\};$$

$m = m_0 = m_1$ (see explanations to formula (1.2.3)); $\Delta t_{f,1}$ and $\Delta t_{f,0}$ - the temperature drop of the hot and the temperature increase of the cold coolants in the regenerator, K; $\bar{t}_{f,j}$ - average values of heat carrier temperatures, K.

The use of formula (1.2.40) is limited to regenerators for blast and open-hearth furnaces with a packing made of low-heat-conducting refractory bricks. In the case of a highly thermally conductive and thin-walled metal packing, the values of the nonstationarity coefficient according to this formula can take negative and imaginary values.

In the work of V.N. Timofeev [26] considers a model of a regenerator for metallurgical furnaces with a brickwork packing. The assumptions adopted by the authors repeat the assumptions of H. Hausen. The authors of [26], considering the essentially conjugate problem of H. Hausen, are looking for a solution to the problem of convective energy transfer in the form of a polynomial:

$$t_{f,j}(\tau) = \sum_{p=0}^{q} \varepsilon_{j,p} \tau^p, \qquad (1.2.41)$$

where j is the period number; $\varepsilon_{j,p}$ –p $(0 \le p \le q)$-th polynomial coefficient for the j-th period.

Then the expression (1.2.41) is substituted into the boundary condition (1.2.29) and the solution of the problem of thermal conductivity of the packing is found as the sum of finite and infinite series in powers of time.

The work [26] is interesting because in it the solution of the adjoint problem is considered as the solution of the following independent problems:

1. From the boundary value problem of the thermal conductivity of the packing, the functional dependence of the temperature of the packing on the transverse coordinate and time is determined under the assumption that the temperatures of the heat carriers in the form of relations (1.2.41) are known.

2. From the equations of convective energy transfer, the temperatures of heat carriers are determined and regression models (1.2.41) are built.

This model is not widely used, since the solution found in [26] has the form of a system of $q + j + 2$ equations, each of which includes several finite and infinite series in powers of τ. As the authors themselves note, "the practical use of the solution is hampered by a large amount of computational work."

Intermediate between analytical and numerical methods for constructing a regenerator model is occupied by the work of A. Lovan [42], in which the temperature fields in coolants and solids are determined by graphical integration of expressions obtained analytically from boundary value problems of the thermal conductivity of a body and coolants.

When building a model of a regenerator, which he mistakenly called a recuperator, Lovan proceeds from two premises:

- the processes in the regenerator of finite length in the first period do not differ from the processes occurring in the regenerator of infinite length;
- the temperature distribution in a real regenerator with periodic contact of the packing with hot and cold heat carriers can be easily obtained from the temperature distribution for an infinite regenerator, in which the initial temperature of the heat carrier is permanently constant.

Therefore, he considers a regenerator with a nozzle in the form of a cylinder of infinite length, washed by a coolant with a constant temperature.

In the formulation part, the Lovan model contains a boundary value problem of one-dimensional thermal conductivity of a cylinder, which consists of a differential equation:

$$\frac{\partial t_w}{\partial \tau} = \lambda_w \left(\frac{\partial^2 t_w}{\partial r^2} + \frac{1}{r} \frac{\partial t_w}{\partial r} \right), \quad \tau > 0 \tag{1.2.42}$$

and boundary condition:

$$\frac{\partial t_w}{\partial r} + \frac{\alpha}{\lambda_w} t_w = \frac{\alpha}{\lambda_w} t_f \quad for \ r = R,$$

as well as the problem of convective energy transfer by a coolant, taking into account the local derivative of the coolant temperature with respect to time:

$$\frac{\partial t_f}{\partial \tau} + w_f \frac{\partial t_f}{\partial y} + \frac{2\pi R\alpha}{S_w \rho_f c_p} t_f = \frac{2\pi R\alpha}{S_w \rho_f c_p} \overline{t}_w, \quad y < w\tau, \tag{1.2.43}$$

with initial condition: $t_{f,0} = 0.$

In equation (1.2.43): R is the outer radius of the cylinder, m; S_w is the cross-sectional area of the cylinder, m^2

Through cumbersome transformations, Lovan obtains a solution in the form of an integral equation of some function in the form of infinite series $2R\alpha / \lambda_w << 6.$ To simplify them, he introduces one more assumption: As a result, the solution for the coolant temperature takes the form:

$$t_f = t_{f,0}\exp(-\xi)\left[\exp\left(-\frac{2\alpha}{R\lambda_w}\eta\right)J_0\left(2\sqrt{\frac{2\alpha\xi}{R\lambda_w}\eta}\right)+\right.$$
$$\left.+\frac{2\alpha}{R\lambda_w}\int_0^{\eta}\exp\left(-\frac{2\alpha}{R\lambda_w}\varsigma\right)J_0\left(2\sqrt{\frac{2\alpha\xi}{R\lambda_w}\varsigma}\right)d\varsigma\right], \quad (1.2.44)$$

Where $\xi = \frac{2\pi R}{S_w\rho_f c_p}y$; $\eta = \lambda_w(\tau - y/w)$.

The temperature of the cylinder is proposed to be determined by the formula

$$t_w = \frac{2\alpha}{R\lambda_w}\sum_{i=1}^{\infty}\exp(-\mu_i^2\tau)\int_0^{\tau}\exp(-\mu_i^2\varsigma)t_f(\xi,\varsigma)d\varsigma \quad (1.2.45)$$

Lovan does not give a complete analytical solution for t_f and t_w by applying the graphical integration of expressions (1.2.44) and (1.2.45).

The statement of the problem in [42] is contradictory: according to the differential equation (1.2.42), the temperature field in the packing is one-dimensional, while the coolant temperature according to equation (1.2.43) changes along the flow.

S.K. Popov [29] constructed a numerical model of a regenerator with a truncated cone-shaped packing from a dense layer of ceramic balls.

§ 1.3. Comparative analysis of known regenerator models

In the practice of thermal calculation of regenerators, the calculation methods (models) of the regenerator, proposed by V.M. Datskovsky [5], S.S. Kutateladze [6], London A. [7], V.K. Blink [8], H. Hausen [11], and A. Shack [12]. These methods were developed at different times under various assumptions and assumptions, which, according to the authors, are the most suitable for one or another regenerator. Therefore, the question arises: to what extent do these models agree with modern concepts of thermal processes in the regenerator and with each other? The answer to this question can be obtained by comparing the results of calculations for different models of the same regenerator.

As an example, a single-section RVP-30 regenerator with a plate packing for a gas turbine plant with a capacity of 4000 kW (GTU-4000), described in [8], is considered. The regenerator has a nozzle made of sheet stainless steel with a thickness of $\delta_w = 0{,}1$mm and a length of $l = 0.1655$ m, placed in a rotor with a diameter of 3 m. The heat carriers are air with a pressure of 0.5 MPa and combustion products of a GTU with a pressure of 0.103 MPa. The temperature of the heat carriers at the inlet is 215 and 425°C. The heat transfer coefficients of air and hot gas are 160.5 and 186.7 W/(m^2•K), respectively. The duration of the heating period of the nozzle is twice the duration of the cooling period.

The models are compared by the nature of the change in the temperature of the packing for a cycle in time, the temperature distribution of the heat carriers and the packing along the length of the regenerator, and by the values of the coefficient of non-stationarity.

The temperature of the packing and coolants was calculated according to the formulas (1.2.4) - the model of S.S. Kutateladze, (1.2.8), (1.2.9) – B.M. Datskovsky, (1.2.10), (1.2.11) – V.K. Blinking and collaborators According to other models, it turned out to be impossible to calculate the indicated temperatures due to the lack of appropriate formulas.

The nonstationarity coefficients were determined by the formulas of S.S. Kutateladze (1.2.6), V.M. Datskovsky, (1.2.9), V.K. Blinking and collaborators (1.2.12), H. Hausen (1.2.39) and A. Shack (1.2.40). According to the formula of A. London-don (1.2.22), the values of thermal efficiency were calculated. The non-stationarity coefficient was then determined by the formula (1.2.23).

The calculation results are shown in Figure 1.1 and 2.2.

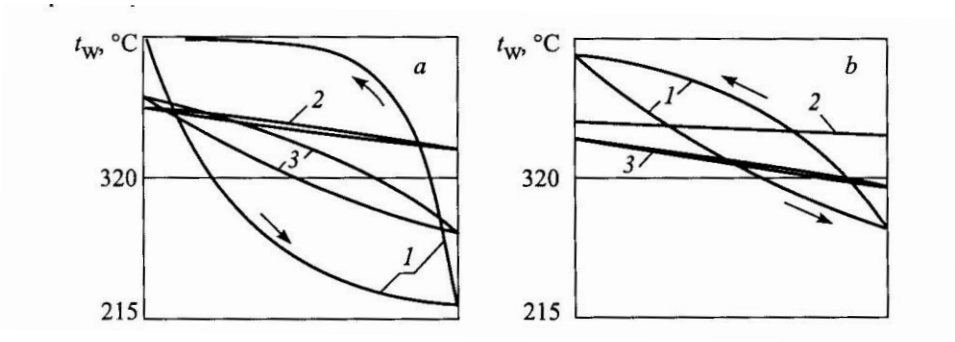


Figure 1.1. Changes in the temperature of the RVP-30 regenerator packing during the cycle. a) n= 5 rpm; b) n = 15 rpm; 1 - according to the Kutateladze model; 2 - according to the Datskovsky model; 3 - according to the Migaya model

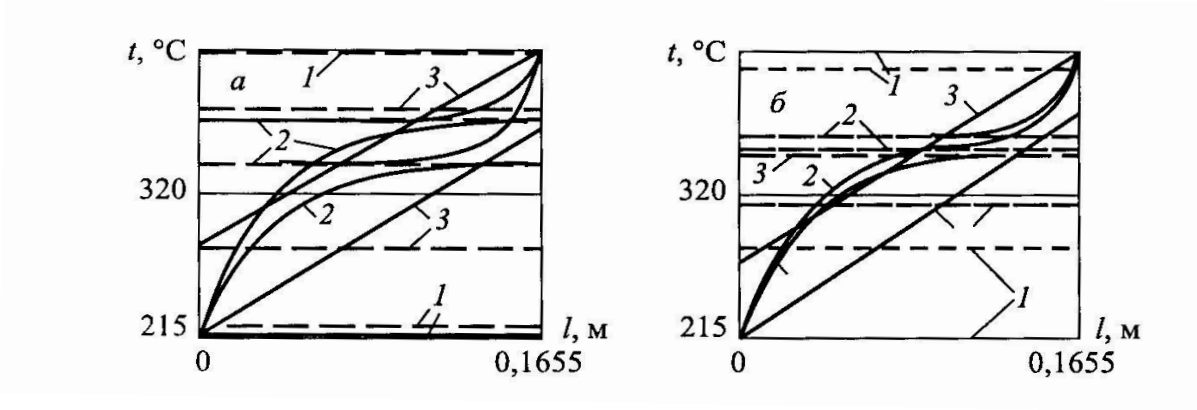


Figure 1.2. Temperature of heat carriers and nozzles of RVP-30 regenerator. Solid curves - temperature of heat carriers; dashed lines – nozzle temperature (see Figure 1.1 for symbols)

Changes per cycle of packing temperatures at the values of cyclic frequency n = 5 and 15 min-1 are shown in Figure2.1. The largest width of the hysteresis loop is obtained according to the Kutateladze model, the smallest - according to Datskovsky; the width of the loop according to the Migaya model takes an intermediate value.

The temperature distribution of heat carriers and packings along the length of the regenerator, shown in Figure 1.2 allows you to install:

1) none of the considered models takes into account changes in packing temperatures along the length of the regenerator (one-dimensional models), which, on the one hand, does not allow determining the location of packing

sections with minimum and maximum allowable temperatures, and on the other hand, contradicts the laws of thermodynamics and the real heat process in the regenerator;

2) the Kutateladze model also neglects changes in the temperature of heat carriers along the flow;

3) according to the Datskovsky model, the temperature of heat carriers over most of the length of the regenerator coincides with the temperature of the packing, which is actually possible only at very high values of the heat transfer coefficients;

4) longitudinal temperature distributions of the packing and coolants according to the Migai model. contradict the second law of thermodynamics: on a significant part of the length of the regenerator during the heating period of the nozzle, its temperature exceeds the temperature of hot gases, and during the cooling period, on the contrary, the temperature of the nozzle is lower than the temperature of cold air.

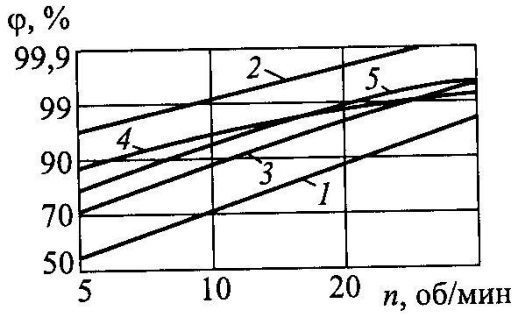


Figure 1.3. Frequency characteristics of RVP-30 by models: 1 – Kutateladze (2.2.6); 2 - Dackovsky (2.2.9); 3 - Flashing (2.2.12); 4 – Hausen (2.2.38); 5 - London (2.2.22), (2.2.45)

The results of calculating the values of the coefficient of non-stationarity in the range of the rotor speed n =5 - 30 rpm are shown in Figure 1.3 in probabilistic-logarithmic coordinates. The figure does not contain the results of calculations using formula (2.2.40), since for a metal nozzle this formula gives negative and imaginary values, which is not consistent with the physical meaning of this quantity. From Figure 1.3 follows:

1) the smallest values of the non-stationarity coefficient are obtained according to the model of S.S. Kutateladze; the largest - according to the model of V.M. Datskovsky (Figure 1.1 and 1.2);

2) discrepancies in the values of the non-stationarity coefficient for different models are 20% at the nominal rotor speed n = 15 rpm; as the frequency decreases, the discrepancy increases and reaches 51% at a frequency of n = 5 rpm.

CHAPTER II. QUALITATIVE INVESTIGATION OF A DISCRETE DYNAMIC SYSTEM MODELING A CYCLIC HEAT TRANSFER PROCESS

This chapter explores a new, simplified mathematical model of the thermodynamic process of heat transfer in a rotating regenerative air heater (in short, an air heater), based on the discretization of both the parameters describing the process itself and the geometric structure of the drum and its rotation. For comparison, §2.1 gives a brief overview of existing models of the mentioned process. Then, in § 2.2, a preliminary, maximally simplified, discrete model is constructed, in § 2.3, the properties of its solutions are established, it is shown that the proposed model allows solving an inverse problem important for practical applications – the problem of determining the thermal conductivity coefficients from measurement results, which is non-linear. In § 2.4, a mathematical model of the heat exchange process in rotating regenerative air heaters in the form of a truncated cone is presented, the advantages of a rotating regenerative air heater in the form of a truncated cone are shown in comparison with existing air heaters.

§ 2.1. The Problem of Mathematical Modeling of the Heat Transfer Process in Rotating Regenerative Air Heaters

As you know, at thermal power plants (abbreviated as TPPs), electricity is generated from thermal energy obtained by burning fuel (gas, coal, fuel oil, etc.). The combustion temperature in the TPP boiler exceeds $500^{o}\,C$ [49]. The release of combustion products with such a temperature into the atmosphere is very wasteful. In order to increase energy efficiency of thermal power plants and reduce thermal pollution of the atmosphere, special units are attached to the station - air heaters, which allow:

a) increase efficiency stations by heating the air blown into the boiler;

b) reduce the temperature of the mixture of gas and smoke emitted into the atmosphere. This process is called regeneration. Modern TPPs use several types

of regenerative air heaters. In principle, a regenerative air heater is a rotating drum (Figure 2.1) stuffed with thin steel flat or corrugated sheets, between which air and hot flue gases pass. (Hereinafter, the mixture of gas and smoke leaving the boiler and emitted into the atmosphere is referred to in one word as “gas”).

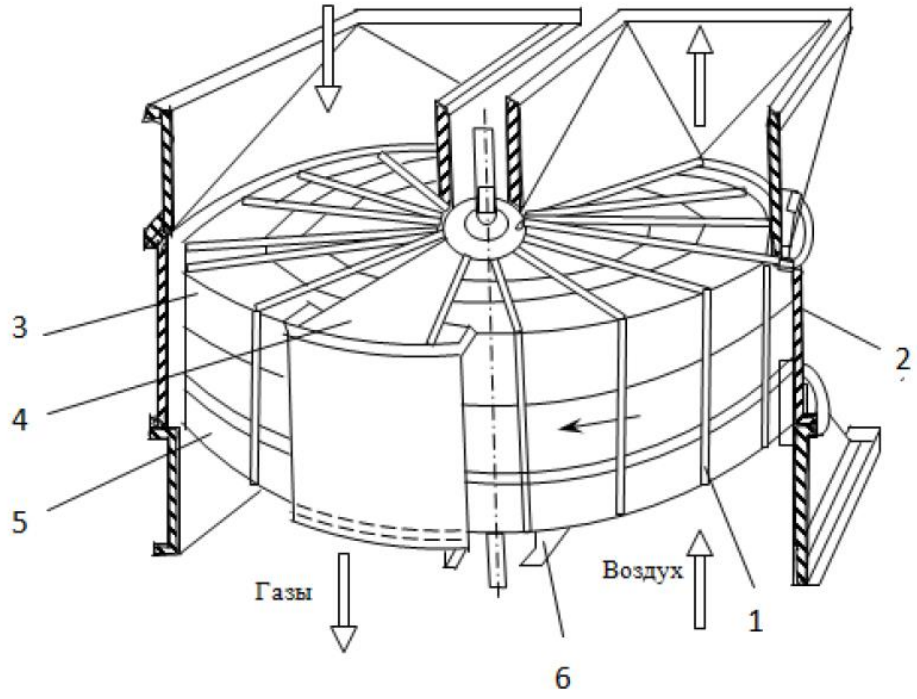


Figure 2.1. General view of the VRVP produced by the company "Jungstrem" (Sweden) [50]: 1 - rotor; 2 - casing; 3 - packages of hot stuffing; 4 – sector plate of radial seal; 5 - packages of cold stuffing; 6 - support beam.

The rotor rotates slowly in a fixed housing. The body is divided into two parts, and the interior of the drum is also conditionally divided into corresponding parts. Gas enters one of them through the neck, air enters the other. The movement of gas and air flows occurs separately in opposite directions. When the rotor rotates, these flows pass through its nozzles, as a result of which the heat of the gas is taken away by the nozzles and given to the air. The process of heat transfer from gas to air is continuous. Heated air is supplied to the boiler of the thermal power plant, increasing its efficiency, and the cooled gas escapes through the pipe into the atmosphere.

Thus, heat transfer is carried out with the participation of three substances: two heat carriers - hot gas to be cooled and cold air to be heated, and metal nozzles of high thermal conductivity [51].

The products of fuel combustion always contain vapors of water and various substances (for example, active sulfur compounds). The condensation temperature (dew point) of such substances can be quite high [52, 53]. Normally, when entering the boiler, the gas temperature is much higher than the dew point, but as it cools, its temperature before leaving the air heater can drop below the dew point, which will lead to the formation of droplets of sulfuric acid and other active chemical compounds that subject the working surfaces of the nozzles to severe corrosion.

In this regard, it is very important to ensure that the temperature regime of the air heater is maintained above the dew point, if necessary, due to additional air heating by means of a heater. When the heater is connected, the temperature of the air supplied from the heater to the air heater rises, which, in turn, leads to an increase in the temperature of the outgoing gas.

This raises two problems:

1. Under what conditions should the heater be turned on?
2. To what temperature is it necessary to heat the air in the heater?

Answers to these questions require knowledge of the temperature distribution of both nozzles and heat carriers in the working volume of the air heater. Obtaining such data by means of measuring instruments is very difficult and expensive. In this regard, the problem of mathematical modeling of the heat exchange process in the air heater arises.

The history of the development of mathematical modeling of the heat transfer process in air heaters dates back to the 1920s, almost simultaneously with the use of air heaters in power plants [51, 54].

To date, various models of the heat exchange process in air heaters have been proposed. A detailed review and comparative analysis of these models are given in [51, 54]. Here we will briefly discuss the main modeling approaches to

make it easier to compare with our approach outlined in the following paragraphs.

In the mathematical models presented in [55, 56, 57, 58], it is assumed that the air heater consists of two areas through which cold and hot coolant flows pass (Figure 2.2).

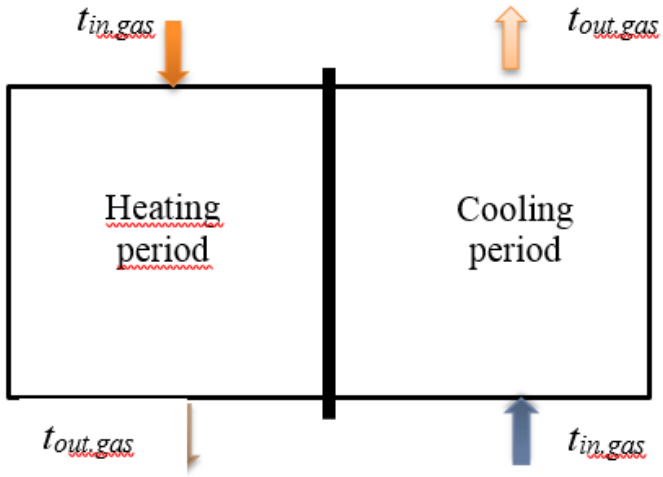


Figure 2.2. Scheme of heat transfer in the air heater.

It is also assumed that the heat exchange process consists of two periods - a heating period and a cooling period. Next, the condition for switching periods in the air heater is formulated, according to which the temperature fields in the nozzle at the end of one period are the initial ones for the next period. Based on these assumptions, equations for the flow energy and heat balance of the packing are taken as a mathematical model:

$$\frac{\partial t_f}{\partial Z} = t_f - t_w; \quad \frac{\partial t_w}{\partial Y} = t_w - t_f, \tag{2.1}$$

where t_w and t_f care the packing and flow temperatures (^{o}C), then the problem is solved with the corresponding initial condition.

Many works are devoted to the study and development of models of this type (for example, see [55, 57, 59, 60]). Although the model is simple and therefore convenient, it can hardly be called adequate to the actual heat exchange process in the air heater.

Among modern analytical models of the heat exchange process in an air heater, the most popular is the mathematical model proposed by Yu.A. Kirsanov [61–52, 54], which is considered quite adequate. Yu.A.Kirsanov considers a single-section air heater with longitudinal nozzles blown by plane-parallel flows of heat carriers (Figure 1.3.) [51, art. 57].

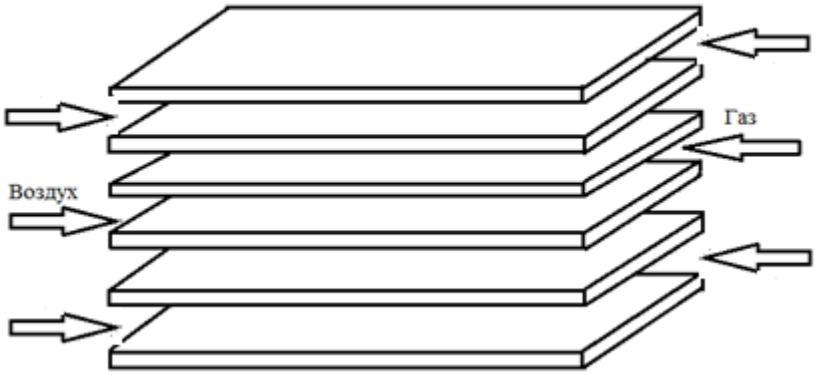


Figure 2.3. Geometry of packings in Yu.A.Kirsanov's model [51].

In Yu. A. Kirsanov, based on the laws of thermodynamics, the process of heat transfer in the nozzles is described by the heat conduction equation

$$\frac{\partial t_{w,j}(x,y,\tau)}{\partial \tau} = a_w \left\{ \frac{\partial}{\partial x}\left[\frac{\partial t_{w,j}(x,y,\tau)}{\partial x} \right] + \frac{\partial^2 t_{w,j}(x,y,\tau)}{\partial y^2} \right\},$$

where $0 \le j \le 3$ is the cycle period number, with the initial condition

$$t_{w,j+1}(x,y,0) = t_{w,j}\left(x,y,T_j\right)$$

and boundary conditions of the third kind

$$\frac{\partial t_{w,j}(0,y,\tau)}{\partial x} = 0,$$

$$\lambda_w \frac{\partial t_{w,j}(\delta_w/2,y,\tau)}{\partial x} = -\alpha_j \left[t_{w,j}(\delta_w/2,y,\tau) - \overline{t}_{f,j}(y,\tau) \right]$$

$$\lambda_w \frac{\partial t_{w,j}(x,0,\tau)}{\partial y} = \alpha_{\text{т},y=0,j} \left[t_{w,j}(x,0,\tau) - \overline{t}_{f,j}(0,\tau) \right],$$

$$\lambda_w \frac{\partial t_{w,j}(x,l,\tau)}{\partial y} = \alpha_{\text{т},y=l,j} \left[t_{w,j}(x,l,\tau) - \overline{t}_{f,j}(l,\tau) \right],$$

as well as the equation of convective energy transfer for a moving medium

$$\rho_f c_f \left[\frac{\partial t_f(x,z,\tau)}{\partial \tau} + w_{f,x} \frac{\partial t_f(z,\varsigma,\tau)}{\partial z} \right] = \lambda_f \left[\frac{\partial^2 t_f(x,z,\tau)}{\partial x^2} + \frac{\partial^2 t_f(x,z,\tau)}{\partial z^2} \right],$$

with initial condition

$$t_f(x,z,0) = \psi(x,z),$$

and boundary conditions

$$\frac{\partial t_f(0,z,\tau)}{\partial x} = 0,$$

$$\frac{\partial t_f(0,z,\tau)}{\partial x} = 0, \quad \lambda_f \frac{\partial t_f(h/2,z,\tau)}{\partial x} = -\alpha_j \left[t_f(h/2,z,\tau) - t_w(\delta_w/2,z,\tau) \right],$$

$$\left\{ \overline{t}_w(z=0,\tau) - \frac{\overline{t}'_f(\tau) + \overline{t}_f(0,\tau)}{2} \right\} = G_f c_{pf} \left[\overline{t}_f(0,\tau) - \overline{t}'_f(\tau) \right],$$

$$\alpha_{\text{т},z=0} F_{\text{т}} \alpha_{\text{т},z=l} F_{\text{т}} \left\{ \overline{t}_w(z=l,\tau) - \frac{\overline{t}''_f(\tau) + \overline{t}_f(l,\tau)}{2} \right\} = G_f c_{pf} \left[\overline{t}''_f(\tau) - \overline{t}_f(l,\tau) \right],$$

$$\overline{t}'(\tau) = f_{\text{вх}}(\tau).$$

where the index f means the parameters of the coolants.

Yu.A. Kirsanov's model has a significant advantage in that it takes into account heat transfer between nozzles and moving heat carrier flows. However, it also has significant drawbacks. So, it does not take into account the rotation of the drum. The geometry of the nozzles in reality is also very different from the system of parallel plates. In addition, it is almost impossible to find an analytical solution, which makes it difficult to use the model in practice.

A review of the literature on mathematical modeling of the heat transfer process in an air heater shows that so far the main approach has been modeling by means of the heat equation. However, such an equation can adequately describe only the process of dynamic heat distribution in the packings; to model the process in coolants - gas and air flows, it is necessary to use the equations of thermogasdynamics, and for heat transfer between packings and coolants, special conjugation relations. Even taking into account these features one has to

deal with other complexities that affect the adequacy of the model. The first of these is that the configuration of the nozzles has a rather complex geometry, due to which the boundary conditions, including the interface conditions on the surface of the nozzles, will be extremely complex. The second difficulty is related to the fact that the flows of heat carriers passing through the air heater drum are not laminar, accompanied by turbulence. An additional complexity is introduced by the need to take into account the rotation of the drum [32].

In [17-19], A. Azamov proposed a new approach to modeling the thermodynamic process of heat transfer in an air heater, based on discretization of both the parameters describing the process itself and the geometric structure of the drum and its rotation. In this case, the model is described by a linear discrete system with a monomial matrix [9], which makes it possible to quite fully investigate the issues of the qualitative behavior of solutions and write out calculation formulas in an explicit form.

§ 2.2. Simplified Discrete Model of Cyclic Heat Transfer Process

Currently, several types of air heaters are used at TPPs. Here we will consider the case of an aggregate whose main block consists of a rotating cylindrical drum with metal nozzles of high thermal conductivity. The area of space occupied by the air heater drum is divided into two parts B_A and B_G a fixed conditional plane passing through the axis of the cylinder. During operation of the air heater B_A, atmospheric air in one direction parallel to the axis of the drum, heating up due to the removal of heat from the nozzles, thereby reducing their temperature. Gas passes through B_G the parts in the opposite direction, cooling due to heat transfer to the nozzles, which are heated at the same time. The final transfer of heat from hot gas to cold air occurs due to the rotation of the drum around its axis at a constant angular velocity. In this section, a mathematical model of the thermodynamic process of an air heater is considered in the form of a simple system of difference equations constructed by averaging the values associated with the process of heat exchange between

nozzles, air and gas both in space coordinates and in time, which will lead to a relatively simple linear discrete equation. The corresponding calculations are given for the case when the averaging over the time variable is performed over a segment equal to the half-period of rotation of the air heater drum. Then the asymptotic properties of the solutions are established. The inverse identification problem is also solved.

2.2.1. Derivation of equations

Let the air heater drum be represented as a cylinder $x^2 + y^2 \le R^2$, $0 \le z \le H$. We will assume that part B_A is determined by the condition $y \ge 0$, and part B_G - by the condition $y \le 0$. Let us denote the temperature at the point in time t, $t \ge 0$, at the point (x, y, z) belonging to the nozzles $\Theta(t, x, y, z)$, and the temperature of the coolant (air or gas) at the point (x, y, z) outside the nozzles - $T(t, x, y, z)$. A couple of quantities $\Theta(t, x, y, z), T(t, x, y, z)$ completely characterize the heat transfer process in the VRVP drum, but, as noted above, due to the complexity of the packing configuration, the initial-boundary problem for the corresponding system of equations of thermo-gas dynamics is difficult to study by analytical methods. One way to overcome this complexity is provided by the averaging method [63, 14]. For this purpose, we consider the corresponding averaged quantities

$$\Theta(t) = \frac{1}{V(B_A^{\bullet})} \int_{B_A^{\bullet}} \Theta(t, x, y, z)\, dv, \quad \Theta_G(t) = \frac{1}{V(B_G^{\bullet})} \int_{B_G^{\bullet}} \Theta_G(t, x, y, z)\, dv,$$

$$T_A(t) = \frac{1}{V(B_A)} \int_{B_A^{\circ}} T_A(t, x, y, z)\, dv, \quad T_G(t) = \frac{1}{V(B_G)} \int_{B_G^{\circ}} T_G(t, x, y, z)\, dv,$$

where $B_A^{\bullet} = B_A \cap \bar{B}$, $B_G^{\bullet} = B_G \cap \bar{B}$, $\bar{B}$ is the part of the drum occupied by nozzles, $B_A^{\circ} = B_A \setminus B_A^{\bullet}$, $B_G^{\circ} = B_G \setminus B_G^{\bullet}$, dv is the volume element, $V(B_A), V(B_G)$ is the volume of parts B_A and B_G, respectively.

Next, we will average and over time intervals $I(n)=\left[nh,(n+1)h\right]$, where h is the half-turn time of the air heater drum $n=0,1,2,\ldots$. The resulting average temperatures are denoted

$$x(n)=\frac{1}{h}\int\limits_{I(n)}\Theta_A(t)dt,\ y(n)=\frac{1}{h}\int\limits_{I(n)}\Theta_G(t)dt,$$

$$u(n)=\frac{1}{h}\int\limits_{I(n)}T_A(t)dt,\ v(n)=\frac{1}{n}\int\limits_{I(n)}T_G(t)dt.$$

We derive the equations relating these quantities under the following simplifying assumptions:

1) When $nh\le t<(n+1)h$ the air heater drum remains motionless, the portion of air (gas) that filled the part B_A° (respectively B_G°) also remains motionless;

2) heat exchange between the coolant and packings occurs in accordance with Newton's linear law [14];

3) at the moment of time $t=(n+1)h$, a heated portion of air with a temperature $u(n)$ leaves the region B_A°, a portion of gas cooled to a temperature $v(n)$ leaves the region B_G°, the drum abruptly turns to 180° (i.e., the material part $B_A^\bullet$ of the drum will transfer to the part $B_G^\bullet$ and vice versa); after that, a part B_A° is filled with a new portion of air from the external environment (or from a heater if it is connected to an air heater, [64, 65, 34]) with a temperature $p(n)$, and a part B_G° is filled with a new portion of gas with a temperature $q(n)$.

In accordance with Newton's law, there are relations

$$\begin{aligned}x(n+1)&=(1-\beta h)y(n)+\beta hq(n),\\ y(n+1)&=(1-\alpha h)x(n)+\alpha hp(n),\end{aligned}\tag{2.2}$$

$$\begin{aligned}u(n)&=p(n)+\gamma h\left(x(n)-p(n)\right),\\ v(n)&=q(n)+\delta h\left(y(n)-q(n)\right).\end{aligned}\tag{2.3}$$

Here, $\alpha,\beta,\gamma,\delta$ are the parameters that depend on the characteristics of the heat transfer process in the HRVP (geometry and heat capacity of the drum

body and nozzle system, composition and humidity of air and gas, their thermodynamic characteristics, thermal conductivity and diffusion coefficients, parameters characterizing the heat exchange process on the contact surfaces of the nozzles with air and gas and etc.).

Note that the pair of equations (2.2) is in form a difference scheme for system (2.1). However, to system (2.1) it is really necessary to add the conjugation condition at the boundaries of the heating and cooling periods. Unlike (2.1), system (2.2) is closed.

2.2.2. Integration of system (2.2) – (2.3)

Introducing the vectors

$$z(n)=\big(x(n),y(n)\big)^T,\ \ r(n)=h\big(\beta q(n),\alpha p(n)\big)^T$$

(where T is the transport sign, the action of which turns the row vector into a column vector) and the matrix

$$A=\begin{pmatrix} 0 & 1-\beta h \\ 1-\alpha h & 0 \end{pmatrix},$$

system (2.2)–(2.3) can be written as

$$z(n+1)=Az(n)+r(n),\ \ n=0,1,2,\ldots. \qquad (2.4)$$

The solution to equation (2.4) is given by the Cauchy formula [66]

$$z(n)=A^n z_0+\sum_{k=0}^{n-1}A^{n-1-k}r(k). \qquad (2.5)$$

Let be $\mu=\sqrt{(1-\alpha h)(1-\beta h)}$. Since $A^n=\mu^n E$, for even n and $A^n=\mu^{n-1}A$ for odd ones n, formula (2.5) can be transformed to the form

$$z(n)=\mu^n z(0)+A\sum_{j=0}^{n/2-1}\mu^{n-2-2j}r(2j)+\sum_{j=0}^{n/2-1}\mu^{n-2-2j}r(2j+1)$$

for even n and

$$z(n)=\mu^{n-1}Az(0)+A\sum_{j=1}^{(n-1)/2}\mu^{n-1-2j}r(2j-1)+\sum_{j=0}^{(n-1)/2}\mu^{n-1-2j}r(2j)$$

for odd ones n.

From here, it is easy to obtain explicit formulas for the quantities $x(n),\ y(n),$ but in practice it is more convenient to calculate them directly from the system of equations (2.2) – (2.3) adapted for computer calculations. In the next section, based on system (2.2) – (2.3), some qualitative properties of solutions are established.

§ 2.3. Qualitative behavior of solutions

2.3.1. Steady-state and periodic regimes

Let us first consider the case of a steady state that arises under the assumption of a constant incoming flow. Such a regime can be established at time intervals measured by hours, in conditions when the air parameters and the energy load at the TPP remain practically unchanged. Due to the fact that it has the order of a minute, the operation of the air heater actually consists of long periods of steady modes and relatively short intervals of transition from one steady state to another. Therefore, it is important to determine the parameters of the air heater in steady-state conditions.

Let us assume that the following inequality holds for the coefficients of system (2.2) – (2.3):

$$0 < \alpha h,\ \beta h < 1, \tag{2.6}$$

which we will call the condition of the *physical feasibility* of the model. This condition leads to the fact that the eigenvalues of the matrix A equal to $\pm\mu = \pm\sqrt{(1-\alpha h)(1-\beta h)}$ lie in the interval $(-1, 1),$ and therefore, all solutions of system (2.4) are asymptotically stable [67].

So, let in system (2.2)–(2.3) $p_n \equiv \overline{p},\ q_n \equiv \overline{q},\ n = 0,1,2,\ldots,\ \overline{r} = (\beta\overline{q},\ \alpha\overline{p})^T.$ Then formula (2.5) simplifies even more: for even n

$$z(n) = \mu^n z(0) + (1-\mu^n)(E-A)^{-1}\overline{r}, \tag{2.7}$$

component by component

$$x(n) = \mu^n x(0) + \frac{1-\mu^n}{\alpha+\beta-\alpha\beta h}[\overline{p} + \overline{q}(1-\beta h)],$$

$$y(n)=\mu^n y(0)+\frac{1-\mu^n}{\alpha+\beta-\alpha\beta h}[\bar{q}+\bar{p}(1-\alpha h)],$$

and for odd n

$$z(n)=\mu^{n-1}Az(0)+\frac{1}{\alpha+\beta-\alpha\beta h}(E-\mu^n A)(E-A)^{-1}\bar{r},$$

component by component

$$x(n)=\mu^{n-1}(1-\beta h)y(0)+\frac{1}{\alpha+\beta-\alpha\beta h}[\bar{p}(1-\mu^{n+2})+\bar{q}(1-\beta h)(1-\mu^{n-1})],$$

$$y(n)=\mu^{n-1}(1-\alpha h)x(0)+\frac{1}{\alpha+\beta-\alpha\beta h}[\bar{q}(1-\mu^{n+2})+\bar{p}(1-\alpha h)(1-\mu^{n-1})].$$

From the equation $\bar{z}=A\bar{z}+\bar{r}$ for the steady state, we obtain $\bar{z}=(E-A)^{-1}\bar{r}$ (the reversibility of the matrix $E-A$ follows from the condition of physical feasibility (2.6)). Component by component

$$\begin{aligned}\bar{x}&=\frac{1}{\alpha+\beta-\alpha\beta h}\left(\alpha\bar{p}+\beta\bar{q}(1-\beta h)\right)\\ \bar{y}&=\frac{1}{\alpha+\beta-\alpha\beta h}\left(\beta\bar{q}+\alpha\bar{p}(1-\alpha h)\right).\end{aligned}\qquad(2.8)$$

By virtue of the above, the steady state regime (2.8) is asymptotically stable.
At time intervals measured in weeks, a periodic regime can be established.

Theorem 2.1. Let the sequence $r(n)$ be periodic with period $m, m\geq 2$, i.e. $r(n+m)=r(n)$, where $n=n_0, n_0+1, n_0+2, \ldots$. Then system (2.4) has a unique periodic solution with period m, which is asymptotically stable.

Proof. Uniqueness and asymptotic stability follow from the physical feasibility condition (2.6). To establish the existence of a periodic regime, we set

$$z_0^*=(E-A^m)^{-1}\sum_{k=0}^{m-1}A^{m-1-k}r_k \qquad(2.9)$$

(matrix invertibility $E-A^m$ is also a consequence of the physical feasibility condition (2.6)). Let $z^*(0)$ be a trajectory with an initial point z_0^*. Relation (2.9) can be rewritten in the form

$$z^*(0) = A^m z_0^* + \sum_{k=0}^{m-1} A^{m-1-k} r(k) = z^*(m).$$

Further,

$$z^*(n+m) = A^{n+m} z_0^* + \sum_{k=0}^{m-1} A^{n+m-1-k} r(k) + \sum_{k=m}^{n+m-1} A^{n+m-1-k} r(k). \qquad (2.10)$$

The first two components of the expression on the right side of (2.10) add up to $A^n z^*(m)$, and the third, due to the periodicity of $r(k)$, is equal to

$$\sum_{k=0}^{n-1} A^{n-1-k} r(k+m) = \sum_{k=0}^{n-1} A^{n-1-k} r(k).$$

But

$$\sum_{k=0}^{n-1} A^{n-1-k} r(k) = z(n) - A^n z(0).$$

Thus, due to (2.10)

$$z^*(n+m) = A^n z^*(m) + z(n) - A^n z_0^* = z(n),$$

Q.E.D.

Corollary 2.1. Let $z(m)$ be finally periodic in the following sense: there exist such N and m that $z(n+m) = z(n)$ for all $n \geq N$. Then system (2.3) has a unique finally periodic solution that is asymptotically stable.

-

At long time intervals under conditions of a variable external environment and a changing energy load on the TPP, the sequence $r(n)$ will not be periodic, much less stationary. In this connection, we present the properties of the solution for more general classes of systems (2.4).

Theorem 2.2. If $r(n)$ is a bounded sequence, then every solution of Eq. (2.4) is also bounded.

This statement is also a special case of more general theorems [67].

Theorem 2.3. Let $\lim_{n\to\infty} r(n) = l$. Then each solution $z(n)$ tends to $(E - A)^{-1} l$ the limit $n \to \infty$ as regardless of z_0.

Proof. Let's first assume $l=0$. Let be $M=\max_n|r(n)|$. For an arbitrary $\varepsilon>0$ we choose such $n_0(\varepsilon)$ that $|r(n)|<\frac{1-\|A\|}{2}\varepsilon$ for $n\ge n_0(\varepsilon)$. Then we choose such $N(n_0(\varepsilon))$ that $\mu^n<\frac{1-\|A\|}{2M\left(1-\|A^{n_0}\|\right)}\varepsilon$, for all $n\ge N(n_0(\varepsilon))$. Obviously, we can choose such $N(n_0(\varepsilon))$ that $N(n_0(\varepsilon))\ge n_0(\varepsilon)$. Then

$$|z(n)|<\|A^n\||z_0|+\left|\sum_{k=0}^{n_0-1}A^{n-1-k}r(k)\right|+\left|\sum_{k=n_0}^{n-1}A^{n-1-k}r(k)\right|.$$

By virtue of condition (2.6), the first term tends to zero as $n\to\infty$. Let us denote the remaining two terms S_n^I and S_n^{II} respectively. Then, taking into account $\|A\|=\mu=\sqrt{(1-\alpha h)(1-\beta h)}\in(0,1)$, we have

$$S_n^I<\|A^{n-n_0}\|\sum_{k=0}^{n_0-1}\|A^{n_0-1-k}\||r(k)|<M\|A^{n-n_0}\|\sum_{k=0}^{n_0-1}\|A^{n_0-1-k}\|<M\mu^n\frac{1-\|A^{n_0}\|}{1-\|A\|}<\frac{\varepsilon}{2}$$

$$S_n^{II}<\sum_{k=n_0}^{n-1}\|A^{n-1-k}\||r(k)|<\frac{1-\|A\|}{2}\varepsilon\sum_{k=0}^{\infty}\|A^k\|=\frac{1-\|A\|}{2}\varepsilon\frac{1}{1-\|A\|}<\frac{\varepsilon}{2}.$$

Hence, $|z(n)|\le S_n^I+S_n^{II}<\varepsilon$ at $n\ge N(n_0(\varepsilon))$, i.e. $z(n)\to 0$.

Let $r(n)\to l$ now for $n\to\infty$, where $l\ne 0$. We make a change of variables $z(n)=\bar z(n)+(E-A)^{-1}l,\ \bar r(n)=r(n)-l.$ Then

$$\bar z(n+1)+(E-A)^{-1}l=A\left[\bar z(n)+(E-A)^{-1}l\right]+\bar r(n)+l.$$

$$\bar z(n+1)=A\bar z(n)+\bar r(n),$$

where $\bar r(n)\to 0$. As proven, $\bar z(n)\to 0$. Hence, $z(n)\to(E-A)^{-1}l.$

In the general case, the change in the parameters characterizing the state of the air, as well as the change in the load on the TPP, is random in nature with a distribution function that is difficult to determine. Here we present another property that takes into account the irregular nature of the magnitude of the phase variables of system (2.2), (2.3).

A sequence a_n is said to be almost periodic if it can be represented as $b_n + c_n$, where b_n is periodic, for $c_n \to 0$ at $n \to \infty$.

Corollary 2.2. If $r(n)$ is almost-periodic, then every solution (2.4) is also almost-periodic.

Definition. A sequence $r(n)$ is called Cesaro summable ((*C*,1) - summable) if the sequence of means

$$\sigma_n = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

converges at $n \to \infty$ [10, 20].

Theorem 2.4. If the sequence $r(n)$ is Cesaro summable, i.e. (*C*,1) is summable, namely,

$$\rho(n) = \frac{r(0) + r(1) + \cdots + r(n-1)}{n} \to l,$$

then each solution $z(n)$ will also be $(C,1)$, summable, and

$$\frac{z(1) + z(2) + \cdots + z(n)}{n} \to (E - A)^{-1} l.$$

Proof. Let be

$$S_N = \frac{1}{N}\sum_{n=1}^{N} z(n) = \frac{1}{N}\sum_{n=1}^{N} A^n z_0 + \frac{1}{N}\sum_{n=1}^{N}\sum_{k=0}^{n-1} A^{n-1-k} r(k).$$

Denoting the terms in the last sum by Σ_I and Σ_{II}, respectively, we have

$$\Sigma_I = \frac{1}{N}\sum_{n=1}^{N} A^n z_0 = \frac{1}{N} A(E - A)^{-1}(E - A^N) z_0 \to 0, \quad \text{at} \quad n \to \infty.$$

Let's calculate the limit of the second sum. We have

$$\Sigma_{II} = \frac{1}{N}\sum_{k=0}^{N-1}\sum_{n=k+1}^{N} A^{n-1-k} r_k = \frac{1}{N}\sum_{k=0}^{N-1}\left(\sum_{n=0}^{N-1-k} A^n\right) r(k) =$$

$$= \frac{(E - A)^{-1}}{N}\left[\sum_{k=0}^{N-1} r(k) - \sum_{k=0}^{N-1} A^{N-k-1} r(k)\right] =$$

$$= (E - A)^{-1}\rho(n) - \sum_{k=0}^{N-1} A^{N-k-1} r(k).$$

By the hypothesis of the theorem, the value of the decrease in the last expression tends to $(E-A)^{-1}l$ as $N\to\infty$. We introduce the notation $\rho(k)=\xi_k+l$, where $\xi_k\to 0$. Then we have

$$r(k)=(k+1)\rho(k+1)-k\rho(k)=(k+1)\xi_{k+1}-k\xi_k+l.$$

Thus,

$$\frac{(E-A)^{-1}}{N}\sum_{k=0}^{N-1}A^{N-k-1}r(k)=\frac{(E-A)^{-1}}{N}\left[\sum_{k=0}^{N-1}A^{N-k-1}\left[(k+1)\xi_{k+1}-k\xi_k\right]-\sum_{k=1}^{N-1}A^{N-k-1}kl\right].$$

Obviously,

$$\sum_{k=0}^{N-1}A^{N-1-k}l\to 0$$

at $N\to\infty$. Let's put

$$S=\frac{1}{N}\sum_{k=0}^{N-1}A^{N-k-1}k\xi_k.$$

We will show $S\to 0$ at $N\to\infty$. Indeed,

$$|S|\le\sum_{k=0}^{N-1}\mu^{N-1-k}\frac{k}{N}|\xi_k|\le\sum_{k=0}^{N-1}\mu^{N-1-k}|\xi_k|.$$

By Theorem 2.3. the right side of this inequality tends to 0 at $N\to\infty$. Thus,

$$S_N=\frac{1}{N}\sum_{n=1}^{N}z(n)\to(E-A)^{-1}l,$$

Q.E.D.

2.3.2. Identification problem for system (2.2) – (2.3)

As noted above, the heat exchange process in the air heater, in addition to the temperature and speed of the incoming air and gas, the speed of rotation of the drum, also depends on many parameters that express the thermodynamic characteristics of air and gas, the material and geometry of the nozzles, the thermal properties of the drum casing, etc. In a number of works devoted to mathematical modeling of air heaters, calculation formulas are given for calculating the values of such parameters based on the laws of molecular physics [51, 57, 1, 4, 5, 11, 17]. At the same time, the determination of the numerical

parameters characterizing the WWTP is possible only with limited accuracy. From this point of view, it is much easier to determine them with satisfactory accuracy by solving the inverse problem for system (2.2)–(2.3) based on empirical data.

As applied to model (2.2)–(2.3), such a problem consists in calculating the values α, β, γ and δ, based on the results of measuring the temperatures of the outgoing air $u(n)$ and gas $v(n)$. Here, we will consider the case when α, β, γ and δ are determined from the measurement results under the assumption $u(1), u(2), v(1), v(2)$ that the incoming flows are stationary: $p(n) = p,\ q(n) = q.$

Thus, we will assume $u(0) = p,\ v(0) = q,$ that the values of $u(1), u(2), v(1)$ and $v(2)$ are found by direct measurement. As a result, we arrive at the problem of finding unknowns $\alpha, \beta, \gamma, \delta$ from the data $h,\ p,\ q,\ u(1), u(2), v(1), v(2),$ avoiding the measurement of quantities $x(n),\ y(n)$ (the temperature of the nozzles). We have

$$x(2) = (1 - \beta h) y(1) + \beta h q, \quad y(2) = (1 - \alpha h) x(1) + \alpha h p, \tag{2.11}$$

$$u(1) = p + \gamma h (x(1) - p), \quad v(1) = q + \delta h (y(1) - q), \tag{2.12}$$

$$u(2) = p + \gamma h (x(2) - p), \quad v(2) = q + \delta h (y(2) - q). \tag{2.13}$$

Let $\bar{u}_k = u(k) - p,\ \bar{v}_k = v(k) - q,\ k = 1, 2,$ (these quantities have a clear physical meaning). After substituting the values $x(2),\ y(2)$ from (2.11) into (2.13), we obtain a non-closed system

$$\gamma h (x(1) - p) = \bar{u}_1, \quad \delta h (y(1) - q) = \bar{v}_1, \tag{2.14}$$

$$\gamma h [(1 - \beta h) y(1) + \beta h q - p] = \bar{u}_2, \quad \delta h [(1 - \alpha h) x(1) + \alpha h p - q] = \bar{v}_2, \tag{2.15}$$

with 6 unknowns $\alpha, \beta, \gamma, \delta,\ x(1),\ y(1)$. It is non-linear and, in general, cannot be solved explicitly. Therefore, we will use the fact that for the values $\alpha, \beta, \gamma, \delta$ there is an a priori estimate $0.01 \div 0.06$ [49], and $h \ll 1$. This allows us to neglect

the terms containing $\beta\gamma h^2$ and $\alpha\delta h^2$, which have order 10^{-4}. Then equations (2.15) take the form $\gamma h(y(1)-p)=\bar{u}_2$, $\delta h(x(1)-q)=\bar{v}_2$. As a result, for intermediate unknowns $x(1)$, $y(1)$ we obtain a linear system

$$\frac{x(1)-p}{y(1)-p}=\frac{\bar{u}_1}{\bar{u}_2}, \quad \frac{y(1)-q}{x(1)-q}=\frac{\bar{v}_2}{\bar{v}_1} \tag{2.16}$$

with a determinant $\bar{u}_1\bar{v}_1-\bar{u}_2\bar{v}_2$. Let's call the value $\chi=\left|\bar{u}_1\bar{v}_1-\bar{u}_2\bar{v}_2\right|$ the discrepancy. The deviation χ from zero is a characteristic of the air heater, expressing the difference in the rates of air heating and gas cooling.

From (2.16) we have

$$x(1)=\frac{(\bar{u}_1-\bar{u}_2)\bar{v}_2p+(\bar{v}_1-\bar{v}_2)\bar{u}_1q}{\bar{u}_1\bar{v}_1-\bar{u}_2\bar{v}_2}, \quad y(1)=\frac{(\bar{u}_1-\bar{u}_2)\bar{v}_1p+(\bar{v}_1-\bar{v}_2)\bar{u}_2q}{\bar{u}_1\bar{v}_1-\bar{u}_2\bar{v}_2} \tag{2.17}$$

Substituting the values $x(1)$, $y(1)$ found from (2.17) into (2.14), (2.15) we obtain the final formulas

$$\alpha h=\frac{x(1)-q}{x(1)-p}\bar{v}_1-\frac{y(1)-q}{x(1)-p}\bar{v}_2, \qquad \beta h=\frac{y(1)-p}{y(1)-q}\bar{u}_1-\frac{x(1)-p}{y(1)-q}\bar{u}_2$$

$$\gamma h=\frac{\bar{u}_1}{x(1)-p}, \quad \delta h=\frac{\bar{v}_1}{y(1)-q}$$

The following table lists the a posteriori values of the parameters $\alpha,\beta,\gamma,\delta$ calculated from these formulas under the assumption $p=32\,^{\circ}\mathrm{C},\ q=282\,^{\circ}\mathrm{C},\ h=0.25$ [54, 20]

Table 2.1

$\bar{u}_1$	$\bar{v}_1$	$\bar{u}_2$	$\bar{v}_1$	α	β	γ	δ
3,61	-2.58	6.75	-4.82	0.048	0.068	0.041	0.029

Comparison with experimental data shows that the found values are close to the real values with an accuracy of $<5\%$.

§2.4. Mathematical model of the heat exchange process in rotating regenerative air heaters in the shape of a truncated cone

Rotating regenerative air heater (RAH) is one of the main types of air heaters used in domestic and foreign power engineering. Despite their wide distribution, their reconstruction and modernization are still relevant, since the existing models of rotating RAH power boilers are not efficient enough. To ensure the transfer of the required amount of heat due to low heat transfer coefficients both on the air and on the gas side, the heat exchange area in the RAH should be large enough. Accordingly, regenerative air heaters are characterized by significant overall dimensions and weight [68]. It is possible to reduce the area through which heat is transferred, to make the installation more compact and lighter, by intensifying heat transfer processes: improving the characteristics of the packing - the heat exchange surface [64, 69] or by using new designs of regenerative air heaters, for example, in the form of a straight truncated cone [64], double-flow, two-way two-stream [69].

In rotating RAHs of traditional cylindrical shape, the cooling of the gas flow (in this case, the specific volume of gases decreases) is accompanied by a decrease in its velocity in the channels of the heat exchange packing, a decrease in the Re number and heat transfer coefficients. Therefore, in the cold part of the regenerative air heater, the heat exchange process is characterized by low heat transfer coefficients. The use of a new design rotating RAH - in the form of a straight truncated cone (Figure 2.4) [64, 69] - will improve heat transfer both in the cold and hot parts of the packing. Reducing the diameter of the regenerative air heater at the outlet of the gas flow and at the inlet of the air flow leads to an increase in the speed of both flows and, accordingly, an increase in heat transfer coefficients and a decrease in the area of the heat exchange surface. As a result, the RAH metal consumption and its cost are reduced.

To determine the operating parameters of a truncated cone RAH, a calculation was performed using a numerical experiment. In accordance with the

research methodology [69, 68], we find the air and gas temperatures at the RAH inlet and outlet from the heat balance equation at given coolant temperatures:

$$Q = G_a c_a (t_a'' - t_a') = G_g c_g (t_g'' - t_g'), \quad (2.4.1)$$

Where Q - the amount of heat given off by the gases and accepted by the air G_a, G_g the air and gas consumption; c_a, c_g – heat capacity of air and gases; t_a', t_a'', t_g', t_g'' - the temperature of air and gases respectively at the inlet to the RAH and the outlet from it.

The speeds of movement of air and gases will be determined taking into account the known areas of the passage sections for gas and air and the average temperatures of the flows:

$$w = G\rho \frac{t + 273}{273 f}, \quad (2.4.2)$$

where G - the coolant flow rate; ρ - the coolant density; t - average coolant temperature; f - cross-sectional area for the passage of the coolant.

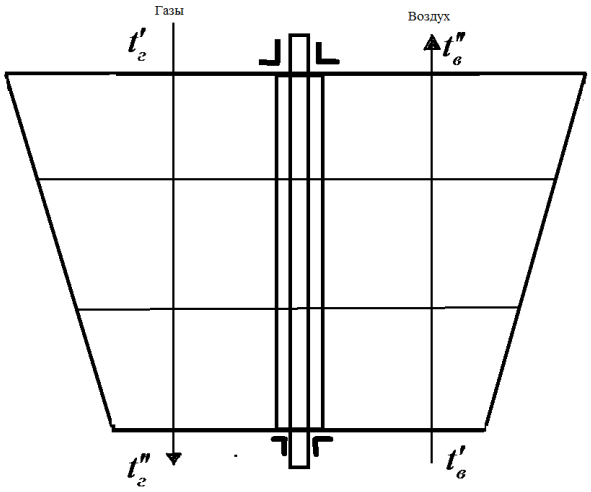


Figure 2.4 Scheme of a single-flow RVP in the form of a truncated cone

According to equation (2.4.2), it is possible to calculate the flow velocities in the RAH of a standard form, based on the constancy of the cross-sectional areas for the passage of air and gas (see Figure 2.5), for the RAH in the form of a truncated cone, the velocities along the height of the packing are taken constant, are determined passage area.

Heat transfer coefficients are calculated by the formula:

$$\alpha = A\frac{\lambda}{d_e}\mathrm{Re}^a\,\mathrm{Pr}^b\,C_t C_l, \qquad (2.4.3)$$

where A - the coefficient determined by the packing type; λ - the thermal conductivity coefficient of the coolant; d_e - equivalent packing diameter; Re - the Reynolds number; Pr - the Prandtl number; a, b - empirical exponents that determine the influence of the criteria Re and Pr on the coefficient α; C_f, C_l - corrections that take into account the influence of the temperature factor and the relative length of the channel.

Based on the known heat transfer coefficients, we determine the heat transfer coefficient:

$$k = \frac{H\xi}{\frac{1}{F_г\alpha_г} + \frac{1}{F_в\alpha_в}}, \qquad (2.4.4)$$

where is a coefficient depending on the RAH rotation speed; H - coefficient of use of RAH packing; ξ - the proportion of the heating surface washed by combustion products or air; α_a, α_g - the heat transfer coefficients from the heat exchange surface to air and from the flue gases to the heat exchange surface.

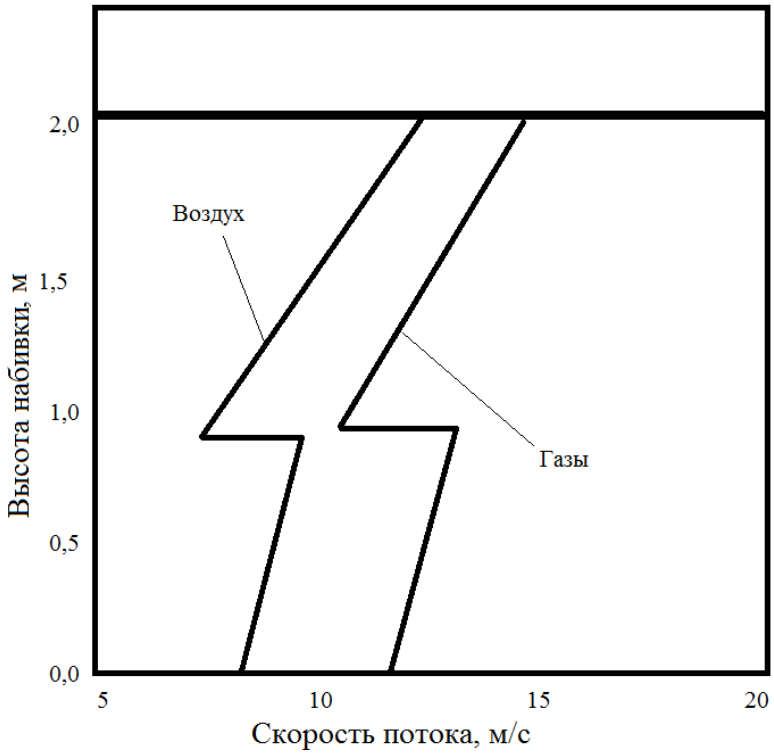


Figure 2.5 Distribution of flow rates in RAH-54 packing height

The area of the heat exchange surface and the height of the RAH packing are calculated from the heat transfer equation:

$$F = \frac{Q}{k\Delta t_{avg}}, \tag{2.4.5}$$

$$H = \frac{F}{F_1} = \frac{Q}{k\Delta t_{avg} F_1}, \tag{2.4.6}$$

where Δt_{avg} the average temperature difference; F_1 - the heating surface enclosed in the packing layer with height $1m$.

Based on this calculation method, a mathematical algorithm and thermal calculation programs "RAH" (for calculating the rotating RAH of an existing design in the form of a cylinder) and "RAH-cone" [70] (for calculating the geometric and thermal characteristics of RAH in the form of a truncated cone) have been developed.

When performing calculations using these programs, the influence of various types of packings on the RAH characteristics is taken into account. When the temperature of heat carriers changes (as it cools or heats up), the thermal properties change (ν - kinematic viscosity, c_p - specific heat capacity, λ - thermal conductivity), which leads to a change in the calculated flow parameters. Therefore, in order to take into account, the influence of a nonlinear change in the properties of heat carriers and for a more detailed calculation, it is proposed to divide the heat exchange surface into small sections along the height and calculate sequentially section by section. The number of iterations depends on the number of small sections and on the value $\Delta t = t_{n+1} - t_n$, where t_n - the flow temperature at the inlet to the section ; t_{n+1} - the temperature of the flow at the inlet to the section $(n+1)$.

Let us assume that the temperature of the flow when moving in the RAH over a short period of time $d\tau$ changes infinitesimally, then the amount of heat $dQ,$ transferred by gases and received by air will be

$$dQ = G_g c_g(t_g) dt_g\,, \tag{2.4.7}$$

$$dQ = G_a c_a(t_a) dt_a\,, \tag{2.4.8}$$

Let us rewrite equation (2.4.6) taking into account expressions (2.4.7) and (2.4.8) in the form of a system of differential equations:

$$\begin{cases} dH = \dfrac{G_g c_g(t_g)}{k(t_g, t_a)\{t_g - t_a\}F_1} dt_g\,, \\ dH = \dfrac{G_a c_a(t_a)}{k(t_g, t_a)\{t_g - t_a\}F_1} dt_a\,. \end{cases} \tag{2.4.9}$$

This system of equations is a mathematical model of heat transfer processes inside the RAH and allows you to determine the required packing height at a given temperature of one of the heat carrier flows.

§2.5. Benefits of a truncated-cone rotary regenerative air preheater compared to existing air preheaters

To establish the operating parameters of rotating cylindrical and conical RAHs, variant calculations were performed. Take as a basis the air heater RAH-54, the most common at the CHPP. Air heaters of this particular type are installed behind power boilers BKZ-420, TGM-84, TGME-464, etc. The initial data for the thermal calculation of RAH-54 are presented below.

Table 2.2

Volumetric consumption of combustion products, m^3/s	59,03
Volume flow rate of air heated in RAH, m^3/s	53,51
Air temperature, ^oC:	
At the entrance to the RAH	30
At the exit from the RAH	220
Temperature of the gases, ^oC:	
At the entrance to the cold stuffing	170
At the exit from the RAH	120
RAH open area, m^2:	
hot part	16,4
cold part	14,96

The results of the thermal calculation of the RAH-54 air heater are as follows:

Table 2.3

	cold stuffing	hot stuffing
Average velocity of gases, *m/s*	12,08	13,25
Average velocity of air, *m/s*	8,67	10,8
Reynolds number for gases	4480	3492

Reynolds number for air	4516	3348
Average heat transfer coefficient, W/($m^2 \cdot K$)	8,75	17,25
Heat dissipation of gases, kW	4095	9938
Heat absorption of air, kW	3890	9441
Heating surface, m^2	4823	7553
Stuffing height, *m*	0,89	1,21

The geometric and thermal parameters of the regenerative air heater in the form of a truncated cone, the prototype of which was RAH-54 are given below:

Table 2.4

	cold stuffing	hot stuffing
Average velocity of gases, *m/s*	16,89	14,82
Average velocity of air, *m/s*	12,4	11,78
Reynolds number for gases	6263	3905
Reynolds number for air	6460	3876
Average heat transfer coefficient, W/($m^2 \cdot K$)	12,88	19,17
Heat dissipation of gases, kW	4095	9938
Heat dissipation of air, kW	3890	9441
Heating surface, m^2	3650	6799
Stuffing height, *m*	0,97	1,13
Rotor inlet diameter, *m*:		
input	4,81	5,4
output	4,54	4,81

The choice of diameters of RAH sections in the form of a truncated cone is based on the fact that along the height of the packing, the speed of flue gases and air is maintained constant and equal to the velocities of these heat carriers at

the entrance to the RAH. When making the final choice of RAH design, it is necessary to perform a feasibility study that takes into account the increase in RAH resistance with an increase in flow velocity and the reduction in capital and operating costs due to a reduction in the volume of heat exchange packing and the total mass of RAH.

Figure 2.6 shows the design of the air heater in the form of a truncated cone with calculated geometric characteristics.

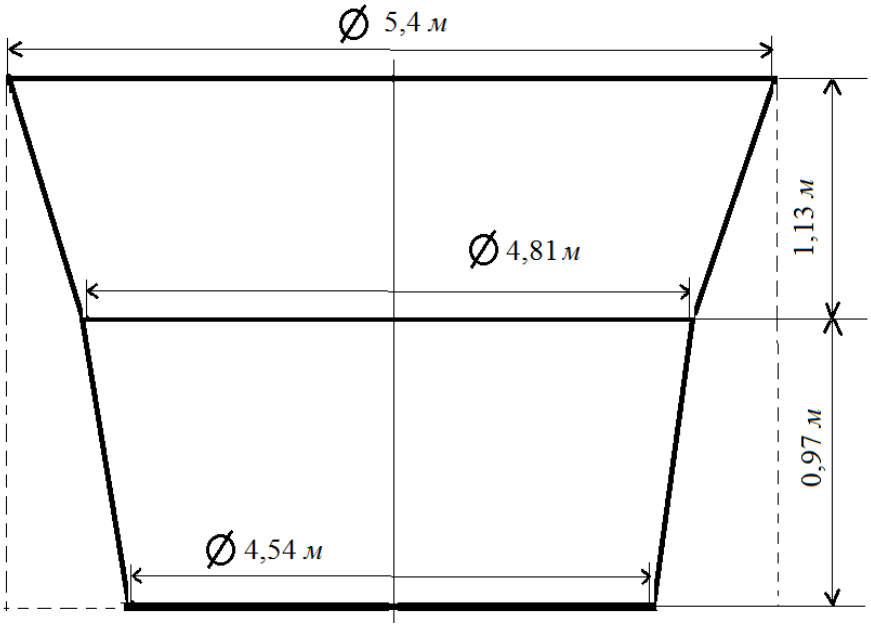


Figure 2.6. The design of the air heater in the form of a truncated cone with geometric calculations.

CHAPTER III. COMPUTER SIMULATION OF THE HEAT TRANSFER PROCESS IN REGENERATIVE HEAT EXCHANGERS

§ 3.1. Modeling Heat Transfer in Regenerative Rotary Air Heaters in COMSOL Multiphysics Finite Element Software

COMSOL Multiphysics is a powerful interactive environment for simulating and calculating most scientific and engineering problems based on finite element differential equations (PDEs). With this software package, we can extend standard differential equation models (applied mode) into multiphysics models to calculate related physical phenomena. The calculation does not require deep knowledge of mathematical physics and the finite element method. This is possible due to the built-in physical modes, where the PDE coefficients are set in the form of understandable physical properties and conditions, such as: heat transfer, heat capacity, heat transfer coefficient, volumetric power, etc. depending on the selected physical partition. The conversion of these parameters into coefficients of mathematical equations occurs automatically. Interaction with the program is possible in a standard way - through a graphical user interface (GUI), or by programming using scripts in COMSOL Script or MATLAB.

The program is based on a system of partial differential equations. There are three mathematical ways to define such systems:

Coefficient form designed for linear and near-linear models

General form, for non-linear models

Weak form, for models with PDEs on boundaries, edges, or for models using conditions with mixed and time derivatives.

Using these methods, we can change the types of analysis, including:

Stationary and transient analysis

Linear and non-linear analysis

Modal and natural frequency analysis

To solve PDE, COMSOL Multiphysics uses the Finite Element Method (FEM). The software runs finite element analysis along with geometry-based meshing and error control using a variety of numerical solvers. Since many physical laws are expressed in the PDE form, it becomes possible to model a wide range of scientific and engineering phenomena from many areas of physics such as: acoustics, chemical reactions, diffusion, electromagnetism, hydrodynamics, filtration, heat transfer, optics, quantum mechanics, semiconductor devices, strength of materials and many others.

In addition to the above, the program allows using coupling variables to connect models in different geometries and connect models of different dimensions to each other.

To create and calculate a task, the following sequence of actions is recommended.

1. Select the dimension of the model, define the physical section in the **Model Navigator** (each section corresponds to a specific differential equation) and define the stationary or non-stationary analysis of the temperature field.

2. Define the work area and set the geometry

3. We set the initial data, the dependence of variables on coordinates and time

4. Specify thermos-physical properties and initial conditions

5. Specify the boundary conditions

6. Set the parameters and build the grid

7. Determine the parameters of the solver and start the calculation.

8. Set the display mode

9. Getting results

In the **COMSOL multiphysics** package, **the Physics>Equation system** command group opens and a partial differential equation is entered that determines the heat transfer process in the RAH. This equation has the form

$$\rho_1 c_g \left(\frac{\partial T_g}{\partial \tau} + w_{x_1} \frac{\partial T_g}{\partial x} + w_{y_1} \frac{\partial T_g}{\partial y} + w_{z_1} \frac{\partial T_g}{\partial z} \right) =$$
$$= \frac{\partial}{\partial x} \lambda_1 \frac{\partial T_g}{\partial x} + \frac{\partial}{\partial y} \lambda_1 \frac{\partial T_g}{\partial y} + \frac{\partial}{\partial z} \lambda_1 \frac{\partial T_g}{\partial z}, \quad (3.1.1)$$

$$\rho_m c_m \frac{\partial T_m}{\partial \tau} = \frac{\partial}{\partial x} \lambda_m \frac{\partial T_m}{\partial x} + \frac{\partial}{\partial y} \lambda_m \frac{\partial T_m}{\partial y} + \frac{\partial}{\partial z} \lambda_m \frac{\partial T_m}{\partial z} \quad (3.1.2)$$

Here, is the number of cycles before the start of the process under study: $\tau_ц$ is the duration of the cycle, equal to the sum of the gas τ_1 and air τ_2 periods, w_{x_1}, w_{y_1}, w_z - are the projections of the gas flow velocity vector in the packing channel, T_g - is the temperature of the "hot" coolant (flue gases), T_m -is the packing temperature, ρ, λ, c_g, c_m - is the density, thermal conductivity coefficient, heat capacity of the hot coolant, heat capacity of the solid body.

With a rectangular channel for heat carriers, the following boundary conditions are set:

initial temperature distribution in a solid

$$T_m(x, y, z, n, \tau_ц) = f_1(x, y, z), \quad (3.1.3)$$

boundary condition of the third kind on the packing surface

$$\lambda_m \frac{\partial T_m}{\partial y}\bigg|_{y=a_1} = \alpha_1(x, a_2, z, \tau)[T_g(x, a_2, z, \tau) - T_m(x, a_2, z, \tau)], \quad (3.1.4)$$

$$-\lambda_m \frac{\partial T_m}{\partial y}\bigg|_{z=a_1} = \alpha_1(x, a_1, y, \tau)[T_g(x, a_1, y, \tau) - T_m(x, a_1, y, \tau)]. \quad (3.1.5)$$

Assuming that the distribution of the flow rates of "hot" and "cold" heat carriers along the channels (RCH) is uniform, one can expect the symmetry of the temperature field. Then in the plane for $z = 0$ and $z = b_1$

$$-\lambda_m \frac{\partial T_m}{\partial z}\bigg|_{z=0} = 0, \quad -\lambda_m = \frac{\partial T_m}{\partial z}\bigg|_{z=b_1} = 0, \quad (3.1.6)$$

planes at $y = 0$ and $y = b_2$

$$-\lambda_m \frac{\partial T_m}{\partial y}\bigg|_{y=0} = 0, \quad -\lambda_m \frac{\partial T_m}{\partial y}\bigg|_{y=b_2} = 0. \tag{3.1.7}$$

Initial temperature distribution in the "hot" coolant at the moment of time $n\tau_ч$

$$T_g(x, y, z, n\tau_ч) = T_{g1} \tag{3.1.8}$$

can be taken equal to the temperature of the "hot" coolant at the inlet to the apparatus.

To add the values of the required coefficients, open the **Physics>Equation system>Subdomain settings** window or **Ctrl+F8**, here you can set the value of any PDE coefficient directly for each Subdomain or change the formula for calculating the coefficients. Each area for which we accept changes will have a lock icon. It will also be visible from the Physics>Subdomain settings window. The priority of all values is in the **Physics>Equation system>Subdomain settings** window, so you need to check which coefficients are already indicated here, and only then set the material properties in **Physics>Subdomain settings**. You can reset all changes to the system of equations using the Reset Equation button.

Table 3.1

Hot gas inlet temperature 623K	623K
Cold air temperature 298K	298K
Hot gas flow rate 7m/s	7m/s
Cold air flow rate 7m/s	7m/s
Rotor rotation speed 2rpm	2rpm
RVP diameter 3m	3m
RVP height 6m	6m

Table 3.2

Material	Density (Kg/m^3)	Thermal conductivity (W/m*K)	Heat capacity KJ/kg*K
Carbon steel	7700	19	510
flue gas	0.616	0.0454	1.047e3
outdoor air	0.580	0.0485	1.05e3

To change the GU formulas, use the **Physics>Equation system>Boundary settings** window or Ctrl+F7. Working with it is similar to the previous one. Using the **Physics>ODE settings** command in COMSOL, you can add any partial differential equations.

3.1.1. Mesh building

After setting all the properties and boundary conditions, the next step is to build the mesh. For the simplest models, at the first stage of the estimated calculation, you can set the default mesh **Mesh>Initialize Mesh** (or the button with the image of a triangle) and press **Mesh>Refine mesh** several times and get a sufficiently fine mesh to start solving. When you click these buttons, the work area switches to **Mesh Mode**, and the mesh is displayed in the work area. This mode can be called manually by the corresponding button  or the **Mesh>Mesh Mode** menu command. For pure conduction models that are not related to mass flow, this can be limited: for smaller mesh elements, the system will automatically refine the mesh, and if it is necessary to further refine the mesh in any part of the system, you can click the **Refine selection** button and specify the desired area. In one-dimensional and two-dimensional stationary mode, it can be advised to simply make the finest grid - the calculation speed on modern computers will still be acceptable. In models that take into account convection, one must always remember that the size of the finite element must be several times smaller than the thickness of the boundary layer, otherwise the

solution may not converge and will be unstable in any case. Therefore, it is recommended to build a grid of such density that there are at least ten finite elements between any two boundaries.

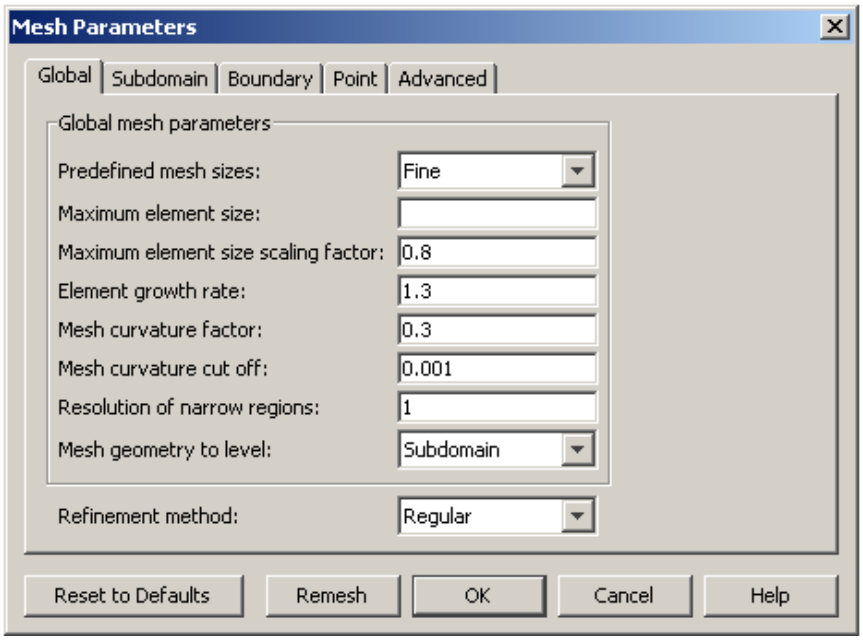


Figure 3.1. Mesh parameters window.

By default, COMSOL generates a triangular mesh in 2D and a tetrahedral mesh in 3D. To set the mesh parameters, select **Mesh>Mesh parameters** or press **F9**. The settings window will open, on the **Global** tab (Figure 3.1) you can select one of the preset modes. There are nine modes in the **Predefined mash sizes** list from **Extremely fine [Extremely accurate]** to **Extremely coarse [Extremely coarse],** the rest are located between these extreme modes. In the fields, you can set your own values for the grid parameters. **Maximum element size** specifies the maximum element size. The default is 1/15 of the maximum side and is optional. **Maximum element size scaling factor** if nothing is set in the previous field, then the value of this field will determine the size of the element (if you set **0.5**, then the element size will be equal to **1/30** of the maximum side, if **0.1** then **1/150**). **Element growth rate [Element growth rate]** is responsible for the degree of condensation, takes values from one to infinity, the closer the value is to one, the more uniform the grid. **Mesh curvature factor** and **Mesh curvature cut off** the smaller these values, the more precisely the curvilinearity of the border is specified: with large values of

these parameters, a broken line will be considered instead of a curve. **Resolution of narrow regions** sets the minimum number of elements along the shortest border; for accurate calculations, it is recommended to set the value of this parameter to at least ten.

The **Mesh geometry to level** field allows you to select separately meshing for regions, surfaces, and edges. For most heat and mass transfer applications, this value should be a **Subdomain**. The **Refinement method** takes two values, **Regular** and **Longest**, and is responsible for the behavior of the **Refine mesh** command. If set to Regular, pressing this command divides each element into four parts in 2D mode, and into eight parts in 3D. Longest value, divides each element into two parts along the longest side. The Subdomain, Boundary, Edge, and Point tabs allow you to set the element size for the corresponding model elements. The Advanced tab allows you to set the mesh anisotropy.

The **Remesh** button rebuilds the mesh with the new parameters.

In 2D mode, for objects close to rectangular, we can set a quadrangular mesh using the menu item **Mesh>Map mesh** or **Ctrl+F9**.

In the Workspace and Geometry section, we mentioned ways to convert 2D models to 3D using the **Draw>Extrude and Draw>Revolve commands**. Then, after setting the three-dimensional geometry, you will have to re-build the mesh of tetrahedra, which can take a lot of time. Sometimes it is advisable to first build a mesh in **2D** mode (triangular or quadrilateral), and then use the **Mesh>Extrude Mesh**… commands to stretch the shape with the mesh or **Mesh>Revolve Mesh**… to unwind the shape with the mesh. Then the elements will not be tetrahedral, but in the form of parallelepipeds or prisms. The time of constructing such a grid is less than building a tetrahedral grid again, but the type of grid does not drastically affect the speed of calculating the problem.

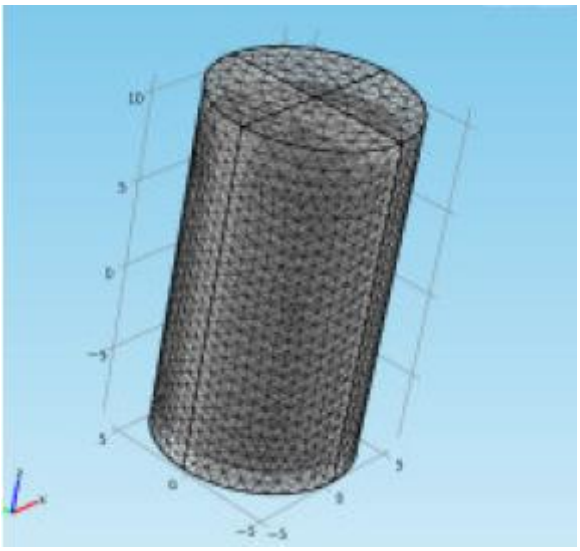


Figure 3.2

Solver. The choice of the decisive device and its parameters is very important, since the reliability of calculations mainly depends on it. Incorrect tuning can lead to gross solution errors that are very difficult to detect. It is also very important to optimize the solution correctly, since, for example, even a not very complex three-dimensional model of a cassette structure element is calculated for tens of minutes on a computer with an Athlon64 3000+ processor and 1Gb of RAM, and some non-linear non-stationary models can be calculated for many hours even on a very powerful computer.

The **Solve** button or the **Solve>Solve problem** menu item launches the solver with the current settings. The **Restart** button or the **Solve>Restart** menu item restarts the solver using the current values (temperature or velocity field) as initial values. If we are considering a stationary problem, then pressing this button should not change the solution. Fluctuations in the values in this case indicate the instability of the solution. It is advisable to use this command for complex calculations, when you can get an approximate solution on a coarse grid and for a linear or stationary solver, and then make a finer grid and, if necessary, change the solver to a nonlinear or transient recalculate the problem. Often this allows you to get a solution faster than immediately running a complex calculation.

To change the parameters, press **F11 Solve>Solver parameters**… or the corresponding button. A window will open (Figure 3.3).

If the **Auto select solver** checkbox is checked, then the program, depending on the application mode, has selected the most suitable device; for simple calculations, most often it is not necessary to change it.

When choosing a decisive device, we must first decide whether we are studying a stationary or transient process. If the process is non-stationary, then in the vast majority of cases, the **Time Dependent** solver is suitable.

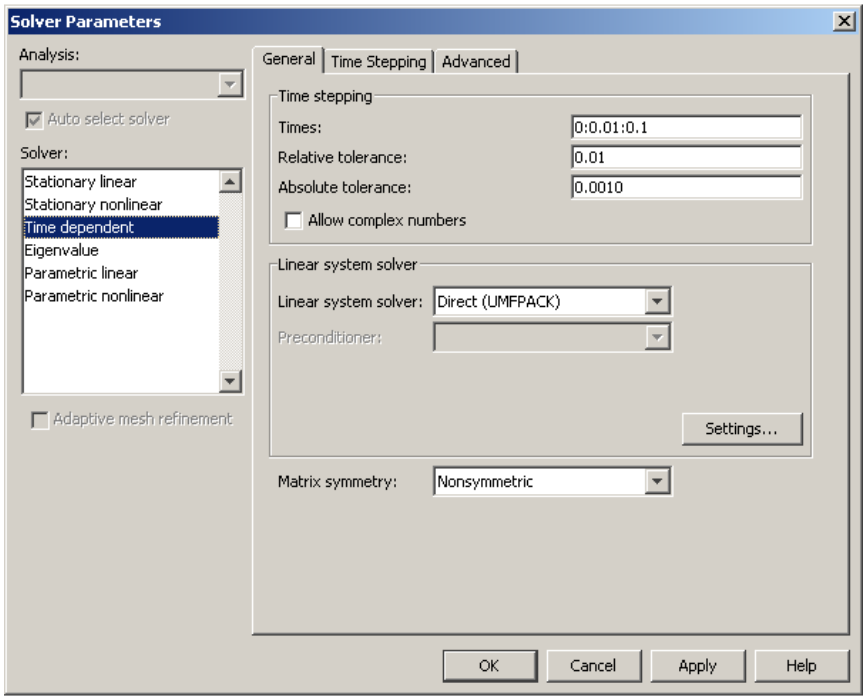
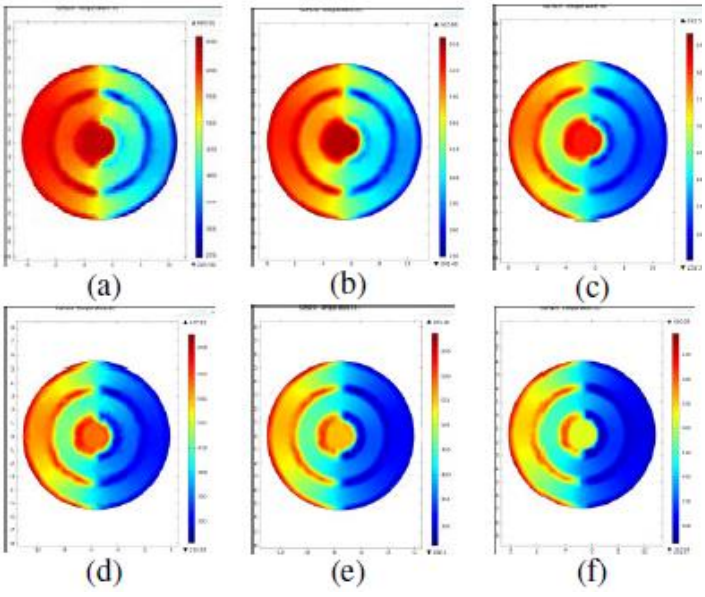


Figure 3.3. Solver parameters window.

3.1.2. Visualization of results

After the completion of the solution, the **Postprocessing mode** is automatically activated, in which you can observe the results of the calculation. This mode can be manually enabled by the corresponding button  or by the command **Postprocessing> Postprocessing mode.**

In a 2D study, the post-processing mode is more extensive. On the left panel are buttons for preset modes. **2D Surface Plot** - displays the value of a scalar field (temperature, velocity or concentration) in color. **3D Surface Plot** - displays the value of a scalar field as a 3D surface. **Contour plot** - displays the solution in the form of isolines (in heat transfer mode - isotherms). **Arrow plot** - displays the vector field (heat flux, velocities) in the form of arrows. **Streamline plot** - displays a vector field in the form of streamlines. **Line Plot** - visualization of values on the borders in color form. **Animate** - Creates an animation of the solution in transition mode. All these buttons are duplicated in the **Postprocessing>Plot Parameters… F12** window, where you can set all kinds of output parameters. On the **General** tab, you can tick off all types of visualization that will be displayed, and set their properties on the tabs. On each tab, you can select any variables or set your own, determine the color match of the results, set the range of output parameters, and much more.



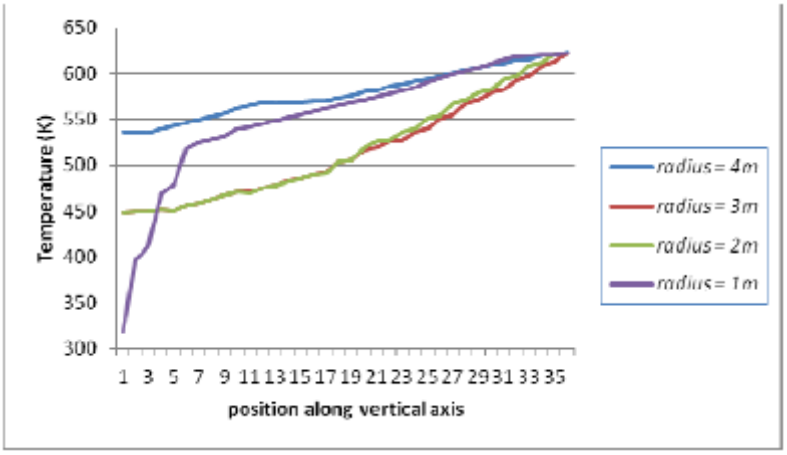


Figure 3.4

To get the value of the displayed value at a specific point, you can click the mouse in the required place of the figure. The value will be displayed in the message line under the workspace along with the coordinates of the point. When the **SNAP** switch is lit, only the values in the grid nodes can be viewed. If you click the **Draw Point for Cross-Section Point Plot**. And then put it on the figure, then a window will open with a graph of temperature changes over time. The **Draw Line for Cross-Section Line Plot** button allows you to draw a straight line through the figure and get a temperature plot along this line. These buttons duplicate the **Postprocessing>Cross-Section Plot Parameters**… menu item, which opens a window with three tabs. On the **General** tab, you can select the time layers or (in the case of a parametric solver) the parameter values for which the graph will be built. The **Point** tab allows you to set the coordinates of the points for which the graph will be built and the variable from which it is built. The **Line** tab also sets the variable and coordinates of the line, it is possible to set the number of equally spaced parallel lines. In the case of transient analysis, a graph will be built for each selected time layer.

The **Domain Plot parameters** menu item allows you to get the type of solution for each area separately or the distribution of the variable on the boundaries and corners of the figure.

In 3D mode, the main visualization type is **Slice Plot**, in this mode a certain number of sections of the computational domain are shown with the distribution of a given variable. **Isosurface Plot** shows isosurfaces. **Subdomain**

Plot shows the distribution of the scalar field of a variable over the entire volume. **Boundary Plot** - shows the temperature distribution at all boundaries of the figure. Other modes are similar to 2D. All parameters of the corresponding imaging modes are configured in the **Postprocessing>Plot Parameters… F12** window

In addition, in the three-dimensional mode, you can see the buttons responsible for "lighting" and the angle of the object.

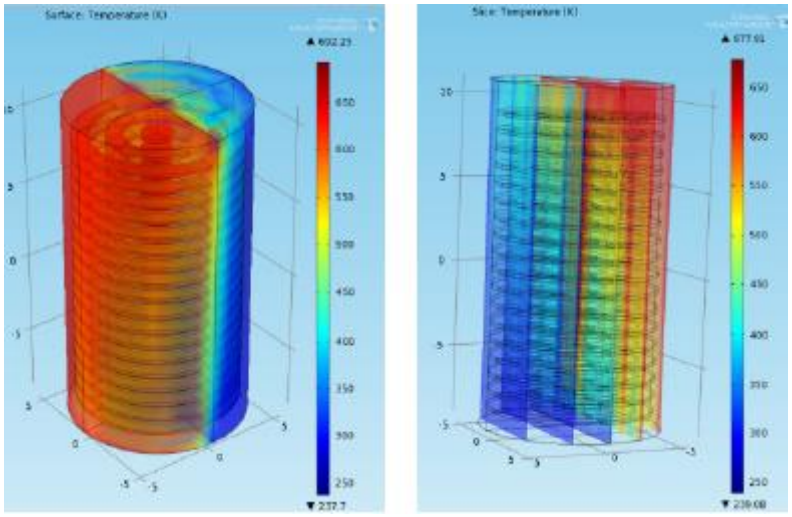


Figure 3.5. Lighting buttons

Often there is a need to integrate some parameter over a volume, surface, or edge. The **Postprocessing>Subdomain/ Boundary/Edge Parameters**… commands allow you to do this: you can select the required element, set a variable or an expression. By the way, to find out the area or volume (for example, to calculate the volumetric power) of an object, you need to set 1 instead of the integrand.

Each of the resulting graphs can be saved both as a picture and as a text file. You can fully export all received data using the **File>Export>Postprocessing** Data menu item.

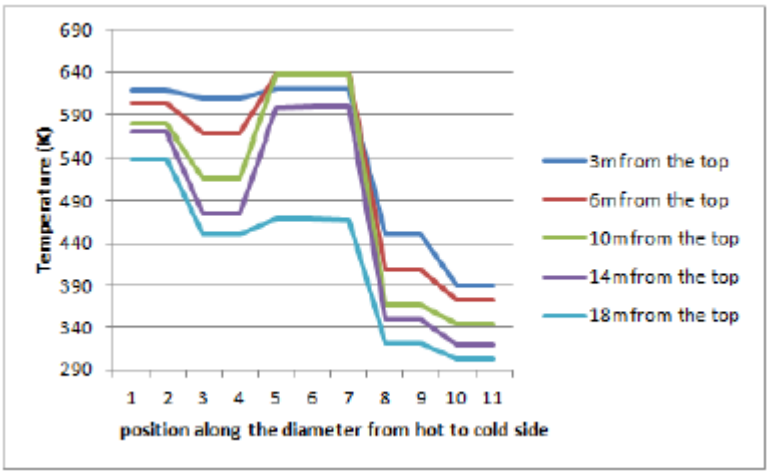


Figure 3.6. A resulting graph to save data

§ 3.2. Computer Program for Numerical Calculation of a Regenerative Air Preheater in Python

On the basis of the mathematical model (2.2) – (2.3), an algorithm was constructed and a program was developed for thermal calculation of the temperature of gases and air leaving the ERVP of thermal power plants in the Python language [35].

The algorithm consists of the following blocks:

Step 1: Enter the number of sectors (m) of the air heater.

Step 2: Enter the initial temperatures of the heat transfer fluids and the air heater nozzle.

Step 3: The coefficients of discrete equations are calculated.

Step 4: First, the temperature of the packing is calculated and then the temperatures of the outgoing cooling gas and hot air are found according to (2.2) - (2.3).

Step 5: The "steady state" conditions will be checked.

The block diagram is shown in Figure 3.7.

Functionality of the program:

- allows you to determine the numerical value of the temperature of the cooling gas and heated air leaving the VRVP;

- the temperature of both incoming and outgoing gas and air is displayed on the animation image;

- temperature graphs are displayed as a function of time;

- allows you to determine the values of the parameters characterizing the thermophysical data (i.e. coefficients $\alpha, \beta, \delta, \gamma$) of the air heater.

The program can be used by scientists engaged in research work in the field of the theory of dynamical systems and its applications, as well as engineers of thermal power plants.

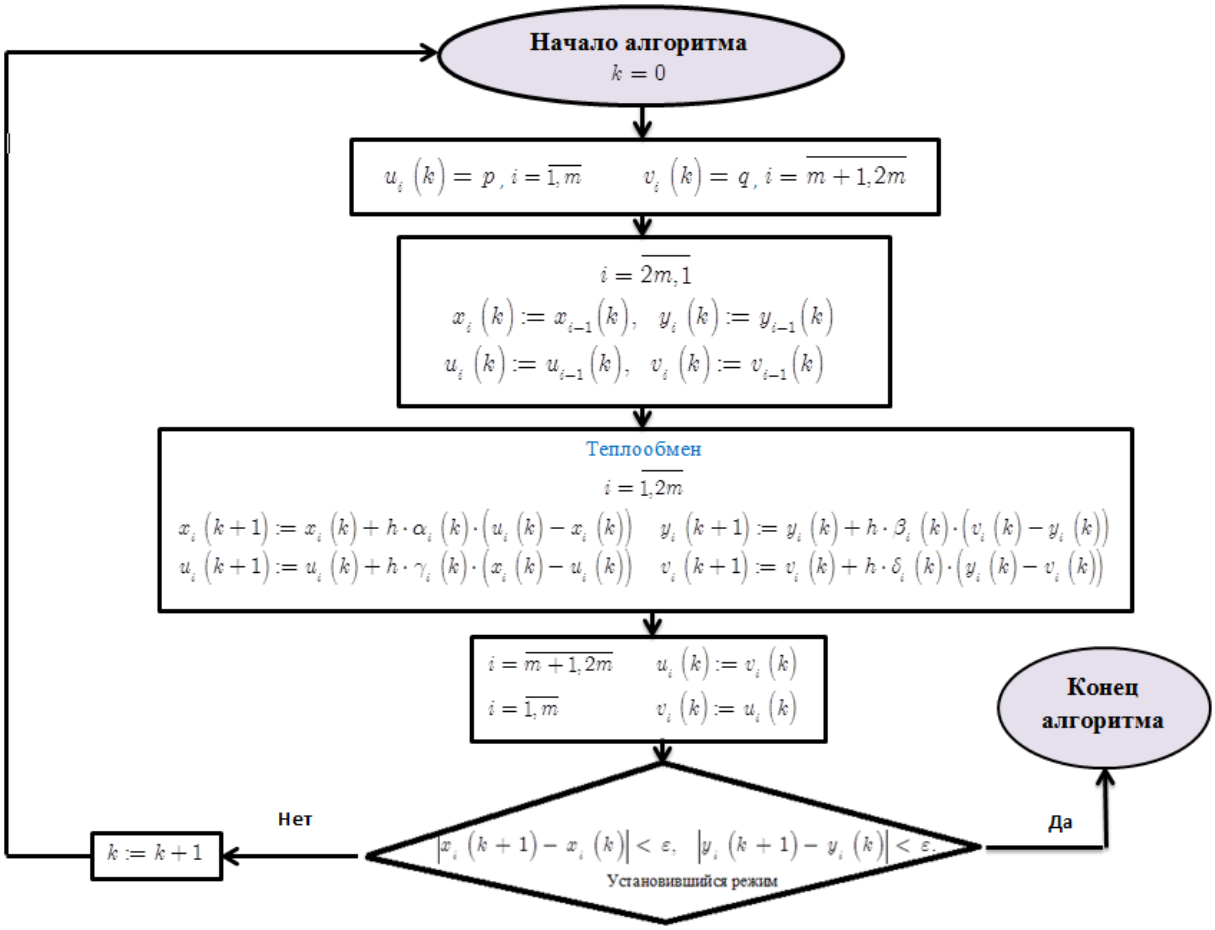


Figure 3.7. Block diagram of the mathematical model of the air heater

Next, we will write a code corresponding to the algorithm above. First, we need to define variables to keep the temperature in each step. To do that we need more advanced type of data structures to keep all data in every procedure. So we have selected not a list but dictionary(dict) as it can receive a tuple in its keys:

```
m = int(input('Enter the number of sectors of the air heater m = '))
k = 0 # initial cycle
x = dict() # temperature at the point (x, y, z) in time t, t >= 0
y = dict() # temperature at the point (x, y, z) in time t, t >= 0
u = dict() # temperatures of the heat transfer fluids
v= dict() # temperatures of the air heater nozzle
x[(0, k)] = 2*m # settled my own
y[(0, k)] = 2*m
u[(0,k)] = int(input('Enter the initial temperatures of the heat transfer fluids u[0] = '))
v[(0,k)] = int(input('Enter the initial temperetures of the air heater nozzle v[0] = '))
p = float(input()) # temperature of gas coming to B_a part
q = float(input()) # temperature of gas coming to B_g part
h = float(input()) # half-turn time of air heater drum  n =0,1,2,...
alpha = float(input())
betta = float(input()) # alpha, betta, gamma, deltas are parameters
gamma = float(input()) # that depend on the characteristics of the heat transfer process in the nozzle
delta = float(input())
epsilon = float(input()) # small quantity needs to be approximated

while True:

    k = 0

    for i in range(1,m+1):
        u[(i, k)] = p
        v[(i+m, k)] = q
```

```
for i in range(2*m, 0, -1):
    x[(i, k)] = x[(i-1, k)]
    y[(i, k)] = y[(i-1, k)]
    u[(i, k)] = u[(i-1, k)]
    v[(i, k)] = v[(i-1, k)]

for i in range(1, 2*m+1):
    x[(i, k+1)] = x[(i, k)] + h*alpha*(u[(i, k)] - x[(i, k)])
    y[(i, k+1)] = y[(i, k)] + h*betta*(v[(i, k)] - y[(i, k)])
    u[(i, k+1)] = u[(i, k)] + h*gamma*(x[(i, k)] - u[(i, k)])
    v[(i, k+1)] = v[(i, k)] + h*delta*(y[(i, k)] - v[(i, k)])

for i in range(m+1, 2*m+1):
    u[(i, k)] = v[(i, k)]
    v[(i - m, k)] = u[(i - m, k)]

if (abs(x[(i, k+1)] - x[(i, k)]) < epsilon) and
        (abs(y[(i, k+1)] - y[(i, k)]) < epsilon):
    print('steady state')
    print(*u)
    print(*v)
    print(*x)
    print(*y)
    break
else:
    k = k+1
```

CONCLUSION

The dissertation work is devoted to the study of dynamic systems simulating cyclic thermal processes.

The main results of the study are as follows.

1. A mathematical model of the heat exchange process of a regenerative rotary air heater has been developed, taking into account geometric structures.

2. The asymptotic properties of discrete dynamic systems are established, which serve as a mathematical model of the thermodynamic process of heat transfer in rotating regenerative air heaters of power plants.

3. The problem of determining the coefficients of the system from the values of the solution of a linear discrete system with a monomial matrix is solved.

4. A computer program for calculating the heat transfer process in rotating regenerative air heaters in the Python language has been developed.

The results obtained can be applied to the study of thermal calculations of regenerative air heaters of thermal power plants. So, in short, it can be said that the results obtained in this dissertation can be further developed and applied to a specific sector of the economy. To do this, we need to further improve the created computer program and compare its numerical indicators with the indicators of the real process, and then, as far as possible, we can bring the program we created closer to the real process. Once we have built the model that is closest to the real process, we will be able to apply it to the application.

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